

Appendix of “Trans-Glasso: A Transfer Learning Approach to Precision Matrix Estimation”

A Comparison of Similarity Assumptions

In this section, we provide a more detailed comparison between our Assumption 1 and the similarity assumption used in [Li et al. \(2023\)](#). Let

$$\Upsilon^{(k)} = (\Omega^{(0)} - \Omega^{(k)})\Sigma^{(k)} = -\Psi^{(k)}\Sigma^{(k)},$$

or equivalently, $\Psi^{(k)} = -\Upsilon^{(k)}\Omega^{(k)}$. [Li et al. \(2023\)](#) assume that $\Upsilon^{(k)}$ is column-wise sparse in the L_p -norm. While our Assumption 1 and the assumption in [Li et al. \(2023\)](#) do not imply each other and are generally not directly comparable, our assumption may be more suitable in certain applications.

First, whereas $\Upsilon^{(k)}$ is motivated by controlling the KL divergence between Gaussian distributions, our Assumption 1 is purely structural and applicable to general distributions. Second, our assumption offers a natural interpretation in Gaussian graphical models, unlike the divergence-based similarity assumption in [Li et al. \(2023\)](#). Finally, a technical advantage of our framework lies in its flexibility when establishing sparsity relationships. While sparsity in $\Psi^{(k)}$ does not necessarily imply sparsity in $\Upsilon^{(k)}$, and vice versa, additional structural assumptions on $\Sigma^{(k)}$ or $\Omega^{(k)}$ can relate the two. Specifically, we have

$$|\Psi^{(k)}|_0 \leq \sum_{i,j} \mathbb{1} \left\{ \sum_{l=1}^d \mathbb{1} \left\{ \Upsilon_{il}^{(k)} \neq 0 \right\} \mathbb{1} \left\{ \Omega_{lj}^{(k)} \neq 0 \right\} \geq 1 \right\}.$$

If both $\Upsilon^{(k)}$ and $\Omega^{(k)}$ are sparse, then $\Psi^{(k)}$ is also sparse. Conversely, if both $\Psi^{(k)}$ and $\Sigma^{(k)}$ are sparse, then $\Upsilon^{(k)}$ must be sparse. A key distinction is that sparsity in $\Upsilon^{(k)}$ implies sparsity in $\Psi^{(k)}$ only when $\Omega^{(k)}$ is sparse, whereas sparsity in $\Psi^{(k)}$ implies sparsity in $\Upsilon^{(k)}$ only when

$\Sigma^{(k)}$ is sparse. In graphical models, sparsity assumptions on the precision matrix $\Omega^{(k)}$ are far more common than on the covariance matrix $\Sigma^{(k)}$, making sparsity in $\Psi^{(k)}$ generally a weaker and more practical assumption than sparsity in $\Upsilon^{(k)}$.

Although a comprehensive theoretical comparison between our Assumption 1 and the similarity assumption in Li et al. (2023) is challenging, we provide some empirical insights in the following discussion. We begin by introducing the divergence distance used in Li et al. (2023) to quantify the difference between $\Omega^{(0)}$ and $\{\Omega^{(k)}\}_{k=1}^K$, defined as

$$h_{\mathcal{D}} := \max_{1 \leq k \leq K} \left\{ \max_{1 \leq j \leq d} \left\| \Upsilon_{j,\cdot}^{(k)} \right\|_p + \max_{1 \leq j \leq d} \left\| \Upsilon_{\cdot,j}^{(k)} \right\|_p \right\},$$

where $\Upsilon_{j,\cdot}^{(k)}$ denotes the j -th row of $\Upsilon^{(k)}$ and $\Upsilon_{\cdot,j}^{(k)}$ denotes its j -th column. Here, $\|\cdot\|_p$ is the L_p vector norm, for which Li et al. (2023) establish theoretical guarantees when $p \in [0, 1]$.

We empirically examine how $h_{\mathcal{D}}$ grows as h increases under the three models discussed in Section 6.1, considering different choices of $p \in [0, 1]$. Specifically, we set $d = 100$ and $K = 5$. For each value of h , we randomly generate 100 sets of target and source precision matrices $\{\Omega_t^{(k)}\}_{k=0}^K$ for $t = 1, \dots, 100$. For each t , we compute $\{\Upsilon_t^{(k)}\}_{k=1}^K$ and the corresponding divergence distance $h_{\mathcal{D},t}$, then estimate the empirical mean and standard error of $\{h_{\mathcal{D},1}, \dots, h_{\mathcal{D},100}\}$, denoted as $\bar{h}_{\mathcal{D}}$ and $\text{SE}_{h_{\mathcal{D}}}$, respectively. We visualize the error bars as $\bar{h}_{\mathcal{D}} \pm \frac{2}{\sqrt{100}} \text{SE}_{h_{\mathcal{D}}}$. Additionally, we report the results for various values of p to examine the effect of different L_p norms.

The results are presented in Figure 1. We observe that the pattern of how $h_{\mathcal{D}}$ increases with h depends on the choice of p , which defines the L_p norm. Notably, when $p = 0$, $h_{\mathcal{D}}$ directly quantifies the sparsity level of the divergence matrices. In this case, even for small values of h , the sparsity of the divergence matrices can be relatively high, aligning with our earlier discussion.

These observations indicate that scenarios can arise where h is small while $h_{\mathcal{D}}$ is large.

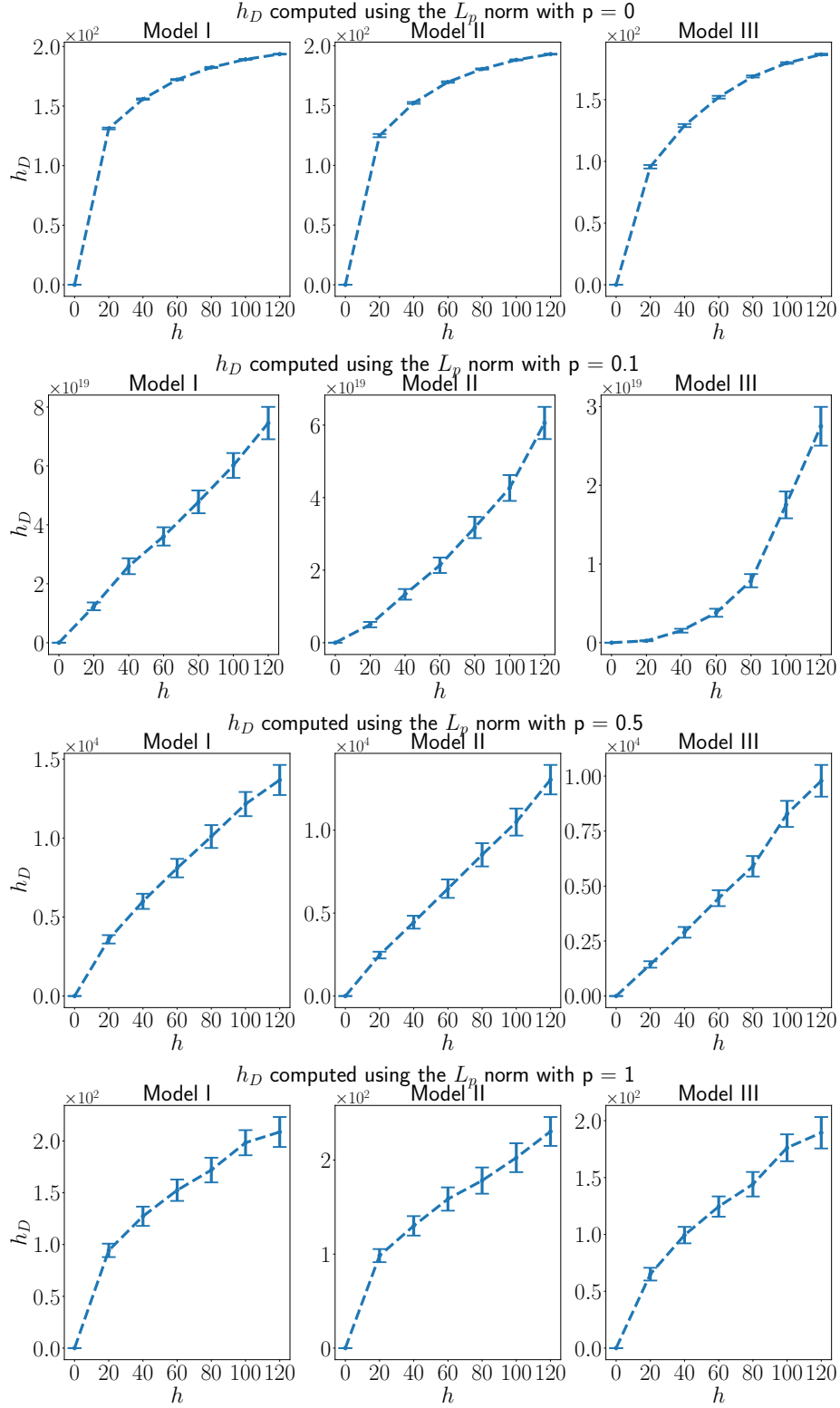


Figure 1: Empirical analysis of how h_D increases with h under the three models described in Section 6.1, for different choices of $p \in [0, 1]$.

This does not imply that Trans-Glasso uniformly outperforms Trans-CLIME; rather, both methods are minimax optimal over different parameter regimes. Our point is that Trans-Glasso can be preferable when h is small but $h_{\mathcal{D}}$ is large, whereas Trans-CLIME may perform better when the opposite holds. In practice, selecting between these similarity assumptions should depend on domain knowledge. A fuller theoretical and empirical comparison of the two approaches is an important direction for future work.

B More Simulation Details

In this section, we provide more simulation details.

For the simulation for Model I, the default setting is $n_0 = 300$, $n_{\text{source}} = 1000$ and $h = 40$. In the first experiment, we increase n_0 while fixing n_{source} and h . In the second experiment, we increase n_{source} while fixing n_0 and h . In the third experiment, we increase both n_0 and n_{source} while increasing h . More specifically, we let $n_{\text{source}} = 3n_0$, and $n_0 = 70$ when $h = 10$, $n_0 = 150$ when $h = 20$, $n_0 = 300$ when $h = 30$, $n_0 = 600$ when $h = 40$ and $n_0 = 1200$ when $h = 50$. In the fourth experiment, we fix both n_0 and n_{source} while increasing h .

For the simulation for Model II, the default setting is $n_0 = 750$, $n_{\text{source}} = 2000$ and $h = 40$. In the first experiment, we increase n_0 while fixing n_{source} and h . In the second experiment, we increase n_{source} while fixing n_0 and h . In the third experiment, we increase both n_0 and n_{source} while increasing h . More specifically, we let $n_{\text{source}} = 3n_0$, and $n_0 = 100$ when $h = 20$, $n_0 = 200$ when $h = 30$, $n_0 = 300$ when $h = 40$, $n_0 = 500$ when $h = 50$, $n_0 = 800$ when $h = 60$, $n_0 = 1000$ when $h = 70$ and $n_0 = 1200$ when $h = 80$. In the fourth experiment, we fix both n_0 and n_{source} while increasing h .

For the simulation for Model III, the default setting is $n_0 = 150$, $n_{\text{source}} = 1000$ and $h = 40$. In the first experiment, we increase n_0 while fixing n_{source} and h . In the second

experiment, we increase n_{source} while fixing n_0 and h . In the third experiment, we increase both n_0 and n_{source} while increasing h . More specifically, we let $n_{\text{source}} = 4n_0$, and $n_0 = 15$ when $h = 20$, $n_0 = 30$ when $h = 30$, $n_0 = 80$ when $h = 40$, $n_0 = 300$ when $h = 50$, and $n_0 = 1000$ when $h = 60$. In the fourth experiment, we fix both n_0 and n_{source} while increasing h .

For the simulation results when the informative set $\mathcal{A}$ is unknown. We set $n_0 = 500$ and $n_{\text{source}} = 1500$ for Model I; $n_0 = 750$ and $n_{\text{source}} = 2500$ for Model II; and $n_0 = 100$ and $n_{\text{source}} = 500$ for Model III.

In Figure 2–5, each dot represents the empirical mean across 30 repetitions, and the vertical bar represents $\text{Mean} \pm \frac{2}{\sqrt{30}} \times \text{Standard Error}$.

C Simulation with Smaller Scales

To investigate the impact of the precision matrix scale on simulation results, we conduct additional experiments in this section. The data are generated using the same setup as Model I in Section 6.1, with a single modification: we set $\tilde{\Omega}_{ij} = 2 \times 0.6^{|i-j|} \mathbb{1}(|i-j| \leq 1)$ for $1 \leq i, j \leq d$, and draw u_{ij} values from $\text{Unif}[-1, 1]$. This adjustment reduces the overall scale of the precision matrix, making it more comparable to the setting examined in the simulation experiments of [Li et al. \(2023\)](#). We refer to this data generation setup as Model IV. The results are presented in Figure 2. As shown, the outcomes closely resemble those of Model I in Figure 2, suggesting that the performance of the various algorithms remains generally robust to changes in the scale of the precision matrix.

D Ablation Study of the Refinement Step

We empirically assess the contribution of the refinement step by evaluating the performance of the initial estimator $\check{\Omega}^{(0)}$, which is the direct output of Trans-MT-Glasso. By comparing

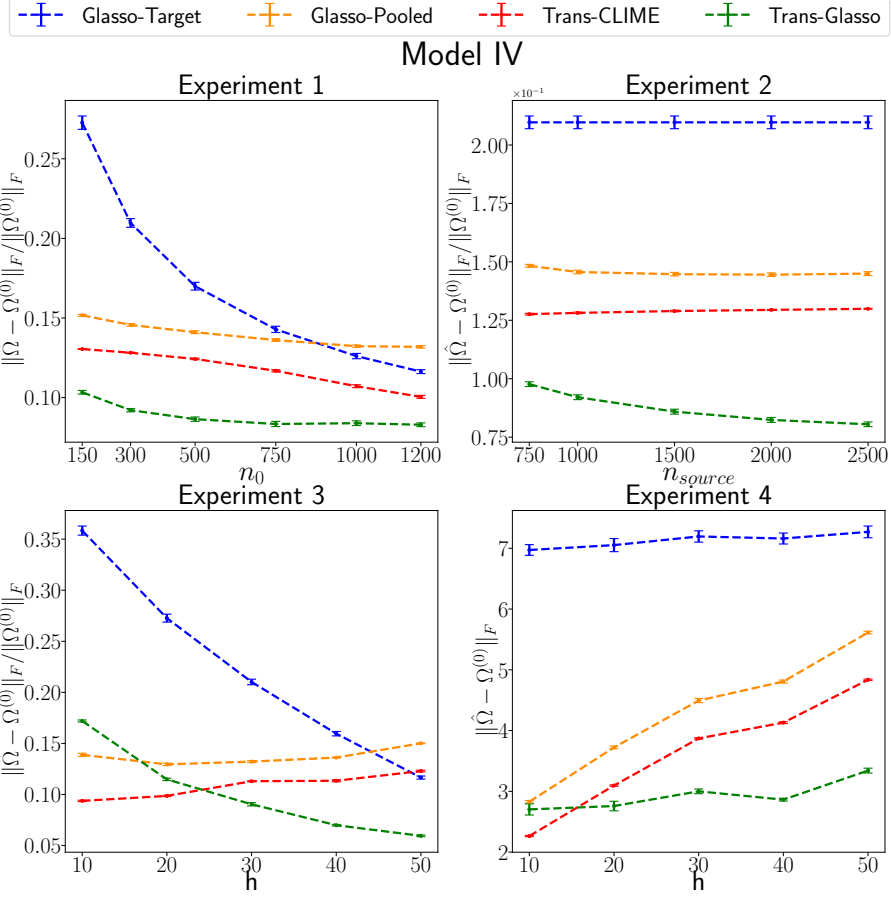


Figure 2: Simulation results for Model IV.

the performance of $\check{\Omega}^{(0)}$ with that of the final estimator $\hat{\Omega}^{(0)}$, we analyze the impact of the refinement step via differential network estimation. We adopt the same simulation settings as in Section 6 and evaluate empirical performance across all three data generation models.

The results are presented in Figures 3–5. As shown, Trans-Glasso consistently outperforms Trans-MT-Glasso, underscoring the crucial role of the refinement step in enhancing estimation accuracy.

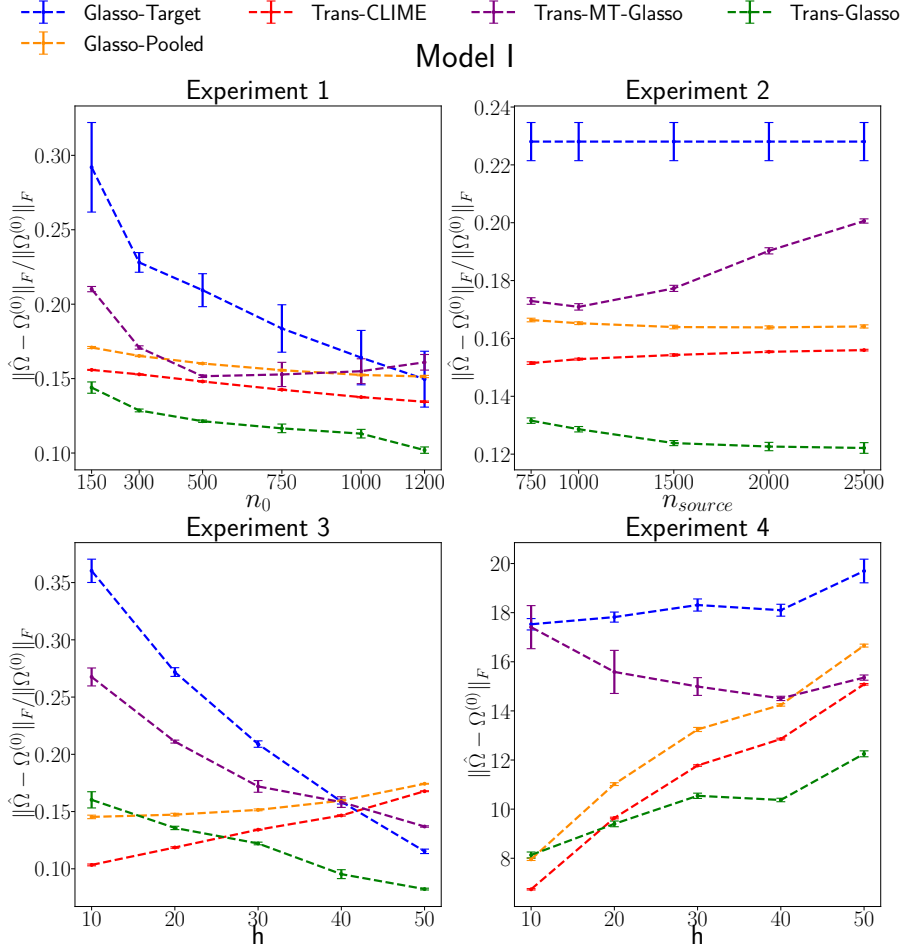


Figure 3: Simulation results of Trans-MT-Glasso for Model I.

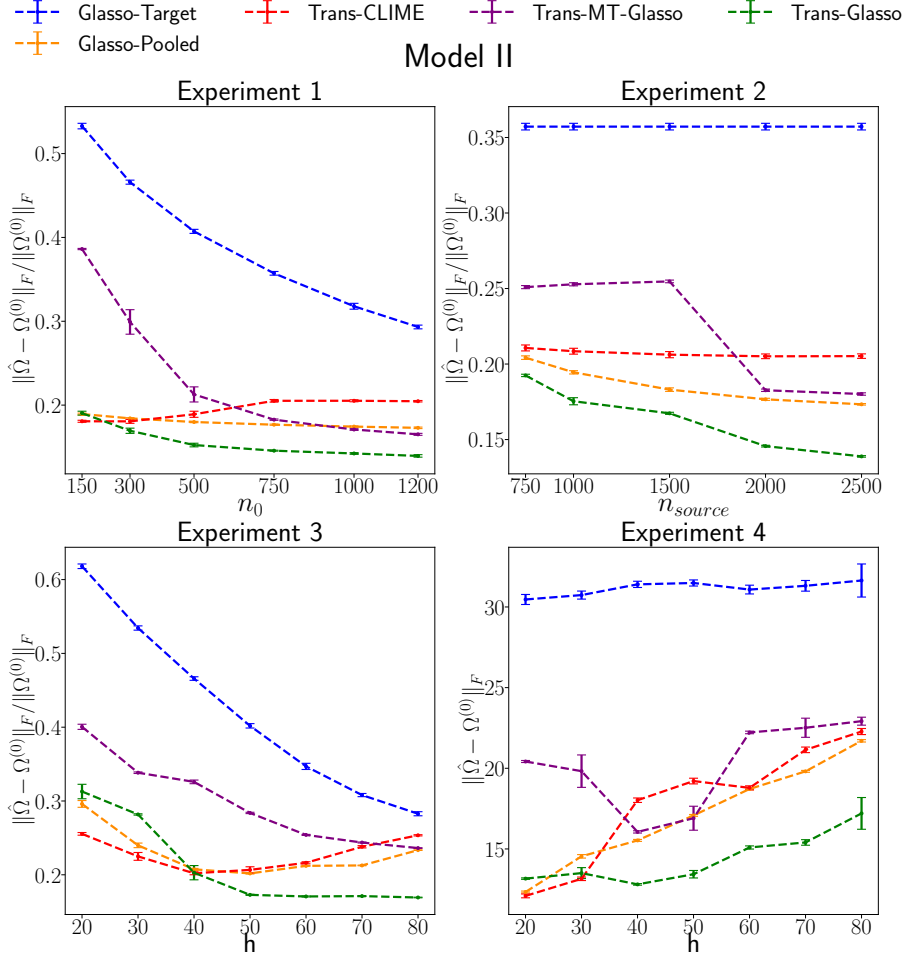


Figure 4: Simulation results of Trans-MT-Glasso for Model II.

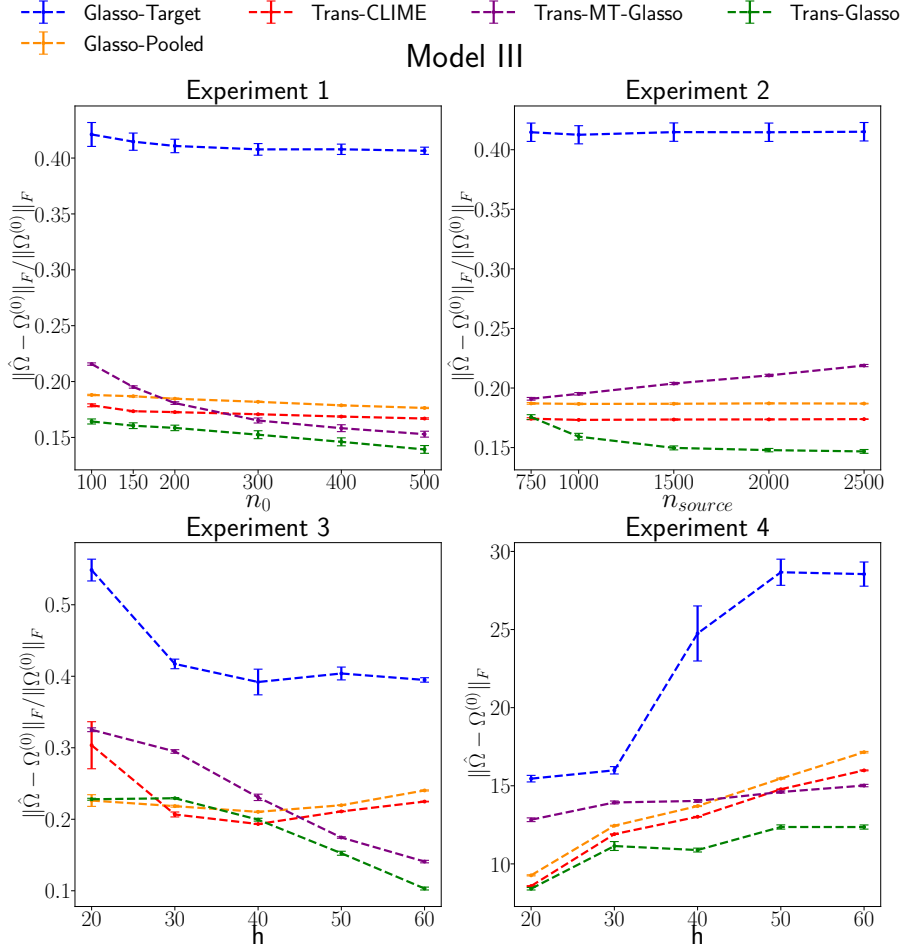


Figure 5: Simulation results of Trans-MT-Glasso for Model III.

E Comparative Analysis of Protein Networks Data Using Trans-CLIME

We apply the Trans-CLIME algorithm proposed by [Li et al. \(2023\)](#) to the protein network data analyzed in Section 7.2. Consistent with the methodology described in Section 7.2, we select the top 20 edges exhibiting the largest absolute values in the estimated precision matrices. The resulting graphs are presented in Figure 1.

Comparing these results to those in Figure 7, we observe that both Trans-Glasso and Trans-CLIME produce identical graphs for subtypes M0 and M2, suggesting a high degree of similarity between these two subtypes in terms of their protein interaction networks. However, the graphs obtained via Trans-CLIME exhibit fewer shared edges overall, indicating that this method tends to highlight greater differences among subtypes. In contrast, the graphs estimated by Trans-Glasso display greater similarity across different subtypes.

This difference in outcomes can be attributed to the distinct assumptions underlying each algorithm. Specifically, Trans-Glasso assumes that most entries in the precision matrices are shared across subtypes, leading to more similar graph structures. In contrast, Trans-CLIME relies on the assumption that the L_p -norm of the divergence matrix is small, which does not necessarily imply structural similarity. Determining which assumption is more biologically relevant to protein interactions requires deeper biological investigation, a question we leave for future work.

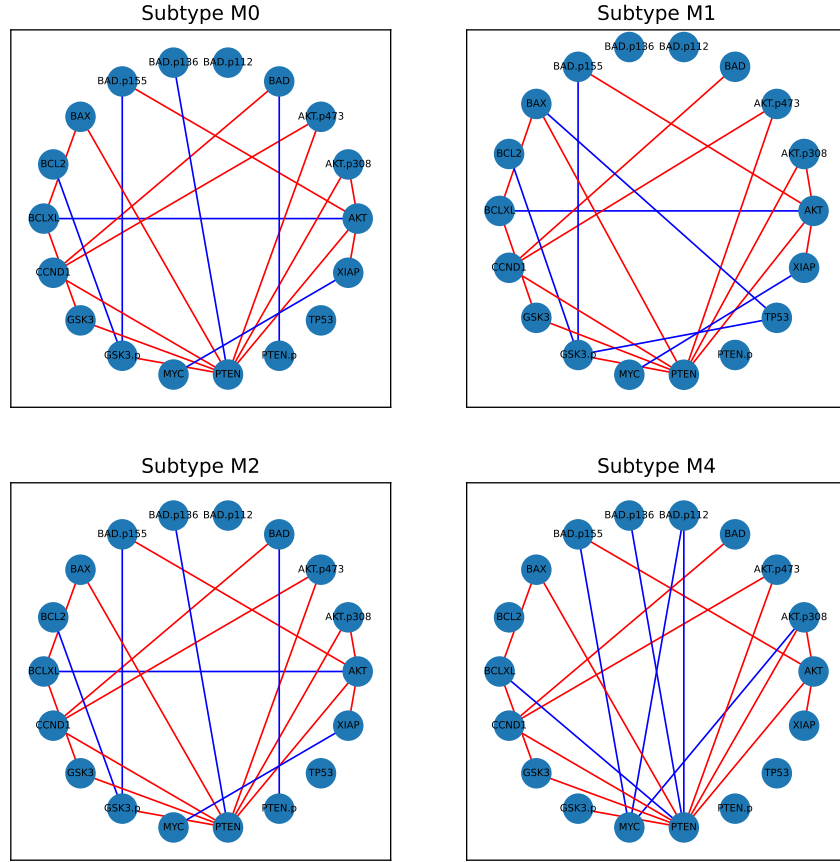


Figure 6: Protein networks for four AML subtypes obtained by Trans-CLIME. Red edges are shared by all subtypes, blue edges are not.

F Preliminary Lemmas

We first collect several inequalities related to matrix norms.

Lemma 1. *For any two matrices $A, B \in \mathbb{R}^{d \times d}$, we have*

$$|AB|_\infty \leq \min \{ \|A\|_\infty \|B\|_\infty, \|B\|_1 \|A\|_\infty \}. \quad (1)$$

Proof. For $1 \leq j \leq d$, denote by $B_{:j}$ the j -th column of B . We then have

$$\begin{aligned} |AB|_\infty &= \max_{1 \leq j \leq d} \|AB_{:j}\|_\infty \\ &\leq \max_{1 \leq j \leq d} \|A\|_\infty \|B_{:j}\|_\infty \\ &= \|A\|_\infty \max_{1 \leq j \leq d} \|B_{:j}\|_\infty \\ &= \|A\|_\infty \|B\|_\infty. \end{aligned}$$

The other claim follows from the facts that $|AB|_\infty = |B^\top A^\top|_\infty$, and $\|B^\top\|_\infty = \|B\|_1$. $\square$

Lemma 2. *For any matrix $A \in \mathbb{R}^{m \times n}$ and any vector $v \in \mathbb{R}^n$, we have $\|Av\|_\infty \leq |A|_\infty \|v\|_1$.*

Proof. Let A_i denote the i -th row of A , we then have

$$\|Av\|_\infty = \max_{1 \leq i \leq m} |\langle A_i, v \rangle| \leq \max_{1 \leq i \leq m} \|A_i\|_\infty \|v\|_1 = |A|_\infty \|v\|_1,$$

which completes the proof. $\square$

The next lemma requires the following definition of the sub-Gaussian random variable.

Definition F.1 (Sub-Gaussian Random Variable, Definition 2 of [Ravikumar et al. \(2011\)](#)).

We say that a random variable $X \in \mathbb{R}$ is sub-Gaussian with parameter σ if

$$\mathbb{E}[\exp(\lambda X)] \leq \exp(\sigma^2 \lambda^2 / 2) \quad \text{for all } \lambda \in \mathbb{R}.$$

Lemma 3. Assume that we obtain samples from K distributions, each with a mean of zero and a covariance matrix of $\Sigma^{(k)}$. Let $\{X_i^{(k)} = (X_{i1}^{(k)}, \dots, X_{id}^{(k)})^\top\}_{i=1}^{n_k}$ represent n_k independently distributed samples from the k -th distribution. In addition, we assume that $X_{ij}^{(k)}/\sqrt{\Sigma_{jj}^{(k)}}$ is sub-Gaussian with parameter σ as defined in Definition F.1. Let

$$\hat{\Sigma}^{(k)} = \frac{1}{n_k} \sum_{i=1}^{n_k} X_i^{(k)} X_i^{(k)\top}, \quad \hat{\Sigma} = \sum_{k=1}^K \alpha_k \hat{\Sigma}^{(k)}, \quad \text{and} \quad \Sigma^* = \sum_{k=1}^K \alpha_k \Sigma^{(k)},$$

where $\alpha_k = n_k/N$ and $N = \sum_{k=1}^K n_k$. For a fixed $\delta \in (0, 1]$, we have

$$\mathbb{P} \left\{ \left| \hat{\Sigma} - \Sigma^* \right|_\infty > 16(1 + 4\sigma^2) \left\{ \max_{1 \leq k \leq K} |\Sigma^{(k)}|_\infty \right\} \max \left\{ \sqrt{\frac{\log(2d^2/\delta)}{2N}}, \frac{\log(2d^2/\delta)}{N} \right\} \right\} \leq \delta.$$

Proof. Let $\bar{X}_{ij}^{(k)} = X_{ij}^{(k)}/\sqrt{\Sigma_{jj}^{(k)}}$, $i = 1, \dots, n_k$, $k = 1, \dots, K$, $j = 1, \dots, d$. Define $U_{i,jl}^{(k)} = \bar{X}_{ij}^{(k)} + \bar{X}_{il}^{(k)}$, $V_{i,jl}^{(k)} = \bar{X}_{ij}^{(k)} - \bar{X}_{il}^{(k)}$, and $\rho_{jl}^{(k)} = \Sigma_{jl}^{(k)}/\sqrt{\Sigma_{jj}^{(k)} \cdot \Sigma_{ll}^{(k)}}$. Then it follows from Proposition 2.9 and (2.18) in [Wainwright \(2019\)](#), that

$$\begin{aligned} \mathbb{E} \left[\exp \left\{ \lambda \left[\left(U_{i,jl}^{(k)} \right)^2 - 2 \left(1 + \rho_{jl}^{(k)} \right) \right] \right\} \right] &\leq \exp \left(\frac{\lambda^2 \nu^2}{2} \right), \quad \text{and} \\ \mathbb{E} \left[\exp \left\{ \lambda \left[\left(V_{i,jl}^{(k)} \right)^2 - 2 \left(1 - \rho_{jl}^{(k)} \right) \right] \right\} \right] &\leq \exp \left(\frac{\lambda^2 \nu^2}{2} \right), \end{aligned}$$

for all $|\lambda| < 1/\phi$ where $\nu = \phi = 16(1 + 4\sigma^2)$. Since

$$\sum_{i=1}^{n_k} \left(\bar{X}_{ij}^{(k)} \bar{X}_{il}^{(k)} - \rho_{jl}^{(k)} \right) = \frac{1}{4} \sum_{i=1}^{n_k} \left(\left(U_{i,jl}^{(k)} \right)^2 - 2 \left(1 + \rho_{jl}^{(k)} \right) \right) - \frac{1}{4} \sum_{i=1}^{n_k} \left(\left(V_{i,jl}^{(k)} \right)^2 - 2 \left(1 - \rho_{jl}^{(k)} \right) \right),$$

we have

$$\begin{aligned} &\mathbb{E} \left[\exp \left\{ \lambda \left[\sum_{i=1}^{n_k} \left(\bar{X}_{ij}^{(k)} \bar{X}_{il}^{(k)} - \rho_{jl}^{(k)} \right) \right] \right\} \right] \\ &= \mathbb{E} \left[\exp \left\{ \frac{\lambda}{4} \left[\sum_{i=1}^{n_k} \left(\left(U_{i,jl}^{(k)} \right)^2 - 2 \left(1 + \rho_{jl}^{(k)} \right) \right) \right] \right\} \cdot \exp \left\{ -\frac{\lambda}{4} \left[\sum_{i=1}^{n_k} \left(\left(V_{i,jl}^{(k)} \right)^2 - 2 \left(1 - \rho_{jl}^{(k)} \right) \right) \right] \right\} \right] \\ &\leq \left\{ \mathbb{E} \left[\exp \left\{ \frac{\lambda}{2} \left[\sum_{i=1}^{n_k} \left(\left(U_{i,jl}^{(k)} \right)^2 - 2 \left(1 + \rho_{jl}^{(k)} \right) \right) \right] \right\} \right] \right\}^{\frac{1}{2}} \times \end{aligned}$$

$$\left\{ \mathbb{E} \left[\exp \left\{ -\frac{\lambda}{2} \left[\sum_{i=1}^{n_k} \left(\left(V_{i,jl}^{(k)} \right)^2 - 2 \left(1 - \rho_{jl}^{(k)} \right) \right) \right] \right\} \right] \right\}^{\frac{1}{2}} \\ \leq \exp \left(\frac{n_k \nu^2 \lambda^2}{8} \right) \quad \text{for all } |\lambda| < \frac{2}{\phi}.$$

Let $\tau_\infty = \max_{1 \leq k \leq K} |\Sigma^{(k)}|_\infty$. It then follows that

$$\begin{aligned} & \mathbb{E} \left[\exp \left\{ \lambda \left[\sum_{i=1}^{n_k} \left(X_{ij}^{(k)} X_{il}^{(k)} - \Sigma_{jl}^{(k)} \right) \right] \right\} \right] \\ &= \mathbb{E} \left[\exp \left\{ \lambda \sqrt{\Sigma_{jj}^{(k)}} \sqrt{\Sigma_{ll}^{(k)}} \left[\sum_{i=1}^{n_k} \left(\bar{X}_{ij}^{(k)} \bar{X}_{il}^{(k)} - \rho_{jl}^{(k)} \right) \right] \right\} \right] \\ &\leq \mathbb{E} \left[\exp \left\{ \lambda |\Sigma^{(k)}|_\infty \left[\sum_{i=1}^{n_k} \left(\bar{X}_{ij}^{(k)} \bar{X}_{il}^{(k)} - \rho_{jl}^{(k)} \right) \right] \right\} \right] \\ &\leq \mathbb{E} \left[\exp \left\{ \lambda \tau_\infty \left[\sum_{i=1}^{n_k} \left(\bar{X}_{ij}^{(k)} \bar{X}_{il}^{(k)} - \rho_{jl}^{(k)} \right) \right] \right\} \right] \\ &\leq \exp \left(\frac{n_k \nu^2 \tau_\infty^2 \lambda^2}{8} \right) \quad \text{for all } |\lambda| < \frac{2}{\phi \tau_\infty}, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E} \left[\exp \left\{ \lambda N \left(\hat{\Sigma}_{jl} - \Sigma_{jl}^\star \right) \right\} \right] &= \mathbb{E} \left[\exp \left\{ \lambda \left(\sum_{k=1}^K \sum_{i=1}^{n_k} \left(X_{ij}^{(k)} X_{il}^{(k)} - \Sigma_{jl}^{(k)} \right) \right) \right\} \right] \\ &= \prod_{k=1}^K \mathbb{E} \left[\exp \left\{ \lambda \left(\sum_{i=1}^{n_k} \left(X_{ij}^{(k)} X_{il}^{(k)} - \Sigma_{jl}^{(k)} \right) \right) \right\} \right] \\ &\leq \prod_{k=1}^K \exp \left(\frac{n_k \nu^2 \tau_\infty^2 \lambda^2}{8} \right) \\ &= \exp \left(\frac{N \nu^2 \tau_\infty^2 \lambda^2}{8} \right) \end{aligned}$$

when $|\lambda| < 2/(\tau_\infty \phi)$. Using Proposition 2.9 in [Wainwright \(2019\)](#), for $t \geq 0$, we have that

$$\mathbb{P} \left\{ \left| \hat{\Sigma}_{jl} - \Sigma_{jl}^\star \right| > t \right\} = \mathbb{P} \left\{ \left| N \left(\hat{\Sigma}_{jl} - \Sigma_{jl}^\star \right) \right| > Nt \right\} \leq 2 \exp \left\{ -\frac{1}{2} \min \left(\frac{4Nt^2}{\nu^2 \tau_\infty^2}, \frac{2Nt}{\phi \tau_\infty} \right) \right\}.$$

A union bound then gives us

$$\mathbb{P} \left\{ \left| \hat{\Sigma} - \Sigma^\star \right|_\infty > t \right\} \leq 2d^2 \exp \left\{ -\frac{1}{2} \min \left(\frac{4Nt^2}{\nu^2 \tau_\infty^2}, \frac{2Nt}{\phi \tau_\infty} \right) \right\}.$$

Rewriting the above equation, we complete the proof. $\square$

Lemma 4. Consider a zero mean random vector $X = (X_1, \dots, X_d)^\top$ with covariance $\Sigma^\star$ such that each $X_j/\sqrt{\Sigma_{jj}^\star}$ is sub-Gaussian with parameter σ as defined in Definition [F.1](#). Let $\{X(i)\}_{i=1}^n$ be n i.i.d. copies of X and let $\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^n X(i)X(i)^\top$. For $\delta \in (0, 1]$, we have

$$\mathbb{P} \left\{ \left| \hat{\Sigma}_n - \Sigma^\star \right|_\infty > 16(1 + 4\sigma^2) |\Sigma^\star|_\infty \max \left\{ \sqrt{\frac{\log(2d^2/\delta)}{2n}}, \frac{\log(2d^2/\delta)}{n} \right\} \right\} \leq \delta.$$

Proof. The proof follows directly from Lemma [3](#). $\square$

In the next lemma, we use $\mathbb{S}_+^d$ to denote the set of positive definite matrices with dimension d .

Lemma 5. Let $A \in \mathbb{S}_+^d$. Define $f(\Omega) := \langle \Omega, A \rangle - \log \det \Omega$ for $\Omega \in \mathbb{S}_+^d$. Given $\Omega_0 \in \mathbb{S}_+^d$ and $\Delta_\Omega \in \mathbb{R}^{d \times d}$ such that $\Omega_0 + \Delta_\Omega \in \mathbb{S}_+^d$, there exists $t \in (0, 1)$ such that

$$f(\Omega_0 + \Delta_\Omega) - f(\Omega_0) - \langle \nabla f(\Omega_0), \Delta_\Omega \rangle = \frac{1}{2} \text{vec}(\Delta_\Omega)^\top \{(\Omega_0 + t\Delta_\Omega)^{-1} \otimes (\Omega_0 + t\Delta_\Omega)^{-1}\} \text{vec}(\Delta_\Omega)$$

Proof. One can view f as a function of $\text{vec}(\Omega)$. Note that $\nabla^2 f(\text{vec}(\Omega)) = \Omega^{-1} \otimes \Omega^{-1}$. The rest of the proof then follows Taylor's Theorem ([Nocedal and Wright, 2006](#), Theorem 2.1). $\square$

Lemma 6. Let $\mathbb{S}^p(s) := \{\theta \in \mathbb{R}^p : \|\theta\|_0 \leq s, \|\theta\|_2 \leq 1\}$. There exists $\{\theta^0, \theta^1, \dots, \theta^M\} \subseteq \mathbb{S}^p(s)$ such that

$$(i) \quad \theta^0 = 0;$$

$$(ii) \quad \|\theta^j - \theta^k\|_2 \geq \frac{1}{4} \text{ for all } 0 \leq j \neq k \leq M;$$

$$(iii) \quad \log(M+1) \geq \frac{s}{2} \log\left(\frac{d-s}{s}\right).$$

Proof. It follows from Example 15.16 in [Wainwright \(2019\)](#) that there exists a $1/2$ -packing of $\mathbb{S}^p(s)$, which we denote as $\{\tilde{\theta}^0, \tilde{\theta}^1, \dots, \tilde{\theta}^M\}$, such that

$$\left\| \tilde{\theta}^j - \tilde{\theta}^k \right\|_2 \geq \frac{1}{2} \quad \text{for all } 0 \leq j \neq k \leq M \quad \text{and} \quad \log(M+1) \geq \frac{s}{2} \log \left(\frac{d-s}{s} \right). \quad (2)$$

Without loss of generality, assume that $\|\tilde{\theta}^0\|_2 \leq \|\tilde{\theta}^j\|_2$ for $1 \leq j \leq M$. Let $\theta^0 = 0$ and let $\theta^j = \tilde{\theta}^j$ for $1 \leq j \leq M$. Then the set $\{\theta^0, \theta^1, \dots, \theta^M\}$ satisfies (i)–(iii). To prove the claim, by our construction of the set and (2), we only need to verify that $\|\theta^j - \theta^0\|_2 = \|\theta^j\|_2 \geq \frac{1}{4}$ for all $1 \leq j \leq M$. We prove the result in two cases.

Case 1: $\|\tilde{\theta}^0\|_2 \geq \frac{1}{4}$. Since $\|\tilde{\theta}^0\|_2 \leq \|\tilde{\theta}^j\|_2$, we have that $\|\theta^j\|_2 = \|\tilde{\theta}^j\|_2 \geq \|\tilde{\theta}^0\|_2 \geq \frac{1}{4}$ for $1 \leq j \leq M$.

Case 2: $\|\tilde{\theta}^0\|_2 < \frac{1}{4}$. We have

$$\|\theta^j\|_2 = \|\tilde{\theta}^j\|_2 \geq \|\tilde{\theta}^j - \tilde{\theta}^0\|_2 - \|\tilde{\theta}^0\|_2 > \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \quad \text{for } 1 \leq j \leq M,$$

where the last inequality follows from (2).

Combining Case 1 and Case 2, we have proved that $\|\theta^j\|_2 \geq \frac{1}{4}$ for $1 \leq j \leq M$, which completes the proof. $\square$

Lemma 7. *Let $m \geq 2$ be an even integer and set $d = 2m$. Let $B \in \mathbb{R}^{m \times m}$ be such that $\|B\|_F \leq 1/4$ and define*

$$\Omega = \begin{bmatrix} I_m & B \\ B^\top & I_m \end{bmatrix}.$$

Then $\Omega > 0$ and $D_{KL}(N(0, \Omega^{-1}) \| N(0, I_d)) \leq \frac{16}{15} \|B\|_F^2$.

Proof. We begin with proving $\Omega > 0$. By Weyl's inequality, we know that

$$\gamma_{\min}(\Omega) \geq 1 - \left\| \begin{bmatrix} 0 & B \\ B^\top & 0 \end{bmatrix} \right\| \geq 1 - \left\| \begin{bmatrix} 0 & B \\ B^\top & 0 \end{bmatrix} \right\|_F$$

$$= 1 - \sqrt{2} \|B\|_F \geq \frac{1}{2}.$$

As a result, we have $\Omega > 0$.

Now we move on to the second claim regarding the KL divergence. Recall that

$$D_{\text{KL}}(N(0, \Omega^{-1}) \| N(0, I_d)) = \frac{1}{2} [\log \det(\Omega) - d + \text{tr}(\Omega^{-1})], \quad (3)$$

which motivates us to compute $\log \det(\Omega)$ and $\text{tr}(\Omega^{-1})$. Let $B = UDV^\top$ be the singular value decomposition of B with $U, D, V \in \mathbb{R}^{m \times m}$, $U^\top U = UU^\top = I_m$, $V^\top V = VV^\top = I_m$, and $D = \text{diag}(\lambda_1, \dots, \lambda_m)$. We have the following identities.

$$\log \det(\Omega) = \sum_{i=1}^m \log(1 - \lambda_i^2). \quad (4a)$$

$$\text{tr}(\Omega^{-1}) = \sum_{i=1}^m \frac{2}{1 - \lambda_i^2}. \quad (4b)$$

Combining (3), (4a), and (4b), we obtain

$$\begin{aligned} D_{\text{KL}}(N(0, \Omega^{-1}) \| N(0, I_d)) &= \frac{1}{2} \left[\sum_{i=1}^m \log(1 - \lambda_i^2) + \sum_{i=1}^m \frac{2}{1 - \lambda_i^2} - d \right] \\ &= \sum_{i=1}^m \left[\frac{\log(1 - \lambda_i^2)}{2} + \frac{1}{1 - \lambda_i^2} - 1 \right]. \end{aligned}$$

Since $\lambda_j^2 \leq \sum_{i=1}^d \lambda_i^2 = \|B\|_F^2 \leq 1/16$, we have $-\lambda_j^2 \geq -1/16$. Also note that $\log(1+x) \leq x$ for all $x \geq -1$. We thus have

$$\begin{aligned} D_{\text{KL}}(N(0, \Omega^{-1}) \| N(0, I_d)) &\leq \sum_{i=1}^m \left[\frac{-\lambda_i^2}{2} + \frac{1}{1 - \lambda_i^2} - 1 \right] \\ &= \sum_{i=1}^m \frac{\lambda_i^2 + \lambda_i^4}{2(1 - \lambda_i^2)} \leq \sum_{i=1}^m \frac{\lambda_i^2}{1 - \lambda_i^2} \leq \frac{16}{15} \sum_{i=1}^m \lambda_i^2 = \frac{16}{15} \|B\|_F^2. \end{aligned}$$

This completes the proof.

Proof of Equation (4a). Using Section 9.1.2 of [Petersen et al. \(2008\)](#), we have $\det(\Omega) = \det(I_m - B^\top B)$. Since

$$I_m - B^\top B = I_m - VD^2V^\top = V(I_m - D^2)V^\top, \quad (5)$$

we have $\det(\Omega) = \det(I_m - D^2) = \prod_{i=1}^m (1 - \lambda_i^2)$ and hence $\log \det(\Omega) = \sum_{i=1}^m \log(1 - \lambda_i^2)$.

Proof of Equation (4b). Using Section 9.1.3 of [Petersen et al. \(2008\)](#), we have

$$\Omega^{-1} = \begin{bmatrix} I_m & B \\ B^\top & I_m \end{bmatrix}^{-1} = \begin{bmatrix} (I_m - BB^\top)^{-1} & -B(I_m - B^\top B)^{-1} \\ - (I_m - B^\top B)^{-1} B^\top & (I_m - B^\top B)^{-1} \end{bmatrix},$$

which implies

$$\text{tr}(\Omega^{-1}) = \text{tr}\{(I_m - BB^\top)^{-1}\} + \text{tr}\{(I_m - B^\top B)^{-1}\}. \quad (6)$$

It follows from (5) that $(I_m - B^\top B)^{-1} = V(I_m - D^2)^{-1}V^\top$, and therefore,

$$\text{tr}\{(I_m - BB^\top)^{-1}\} = \text{tr}\{V(I_m - D^2)^{-1}V^\top\} = \text{tr}\{(I_m - D^2)^{-1}\} = \sum_{i=1}^m \frac{1}{1 - \lambda_i^2}. \quad (7)$$

Similarly, we have $\text{tr}\{(I_m - B^\top B)^{-1}\} = \sum_{i=1}^m \frac{1}{1 - \lambda_i^2}$. Take the previous results collectively to yield the claim (4b). $\square$

G Proof of Theorem 1

We adopt the proof strategy laid out in [Wainwright \(2019, Chapter 9\)](#).

G.1 Additional Notation

We first introduce some additional notation that will be helpful in the proof. We define the Hilbert space for the parameters

$$\mathbb{H} := \left\{ \Theta = \left(\Omega, \{\Gamma^{(k)}\}_{k=0}^K \right) : \Omega, \Gamma^{(k)} \in \mathbb{R}^{d \times d} \text{ for all } 0 \leq k \leq K \right\},$$

with the associated inner product

$$\langle \Theta, \Theta' \rangle_{\mathbb{H}} = \langle \Theta, \Theta' \rangle := \langle \Omega, \Omega' \rangle + \sum_{k=0}^K \langle \Gamma^{(k)}, \Gamma^{(k)'} \rangle.$$

The space $\mathbb{H}$ endowed with the inner product $\langle \cdot, \cdot \rangle$ is indeed a Hilbert space. Correspondingly, we have

$$\|\Theta\|_{\mathbb{H}}^2 = \|\Omega\|_{\mathbb{F}}^2 + \sum_{k=0}^K \|\Gamma^{(k)}\|_{\mathbb{F}}^2.$$

We also need the dual norm of $\Phi(\cdot)$, which is defined as

$$\Phi^*(\Theta) := \sup_{\Theta': \Phi(\Theta') \leq 1} \langle \Theta, \Theta' \rangle. \quad (8)$$

Recall that $\Theta^* = (\Omega^*, \{\Gamma^{(k)*}\}_{k=0}^K)$ is the true parameter. Let $\mathcal{S}_{\Omega}$ and $\mathcal{S}_{\Gamma^{(k)}}$ be the supports of Ω^* and $\Gamma^{(k)*}$ for all $0 \leq k \leq K$, respectively. Under Assumption 1, the true parameter Θ^* lies in the following subspace of $\mathbb{H}$:

$$\mathbb{M} := \left\{ \Theta = \left(\Omega, \{\Gamma^{(k)}\}_{k=0}^K \right) : \text{supp}(\Omega) \subseteq \mathcal{S}_{\Omega}, \text{supp}[\Gamma^{(k)}] \subseteq \mathcal{S}_{\Gamma^{(k)}} \text{ for all } 0 \leq k \leq K \right\}. \quad (9)$$

The orthogonal complement of $\mathbb{M}$ is given by

$$\mathbb{M}^{\perp} := \left\{ \Theta = \left(\Omega, \{\Gamma^{(k)}\}_{k=0}^K \right) : \text{supp}(\Omega) \subseteq \mathcal{S}_{\Omega}^c, \text{supp}[\Gamma^{(k)}] \subseteq \mathcal{S}_{\Gamma^{(k)}}^c \text{ for all } 0 \leq k \leq K \right\}. \quad (10)$$

Clearly, for any $\Theta \in \mathbb{M}$ and $\Theta' \in \mathbb{M}^{\perp}$, we have that $\langle \Theta, \Theta' \rangle = 0$.

We also need to define the projection of a parameter onto a subspace. For any matrix $B \in \mathbb{R}^{d \times d}$ and any subspace $\mathcal{F} \subseteq \mathbb{R}^{d \times d}$, we define

$$[B]_{\mathcal{F}} = \arg \min_{\tilde{B} \in \mathcal{F}} \|\tilde{B} - B\|_{\mathbb{F}}.$$

Similarly, for any $\Theta \in \mathbb{H}$ and any subspace $\mathbb{F} \subseteq \mathbb{H}$, we define

$$[\Theta]_{\mathbb{F}} = \arg \min_{\tilde{B} \in \mathbb{F}} \|\tilde{B} - \Theta\|_{\mathbb{H}}.$$

In addition, for $S \subseteq [d] \times [d]$, we define

$$\mathcal{M}(S) := \{B \in \mathbb{R}^{d \times d} : \text{supp}(B) \subseteq S\}.$$

For any $B \in \mathbb{R}^{d \times d}$, we define $[B]_S := [B]_{\mathcal{M}(S)}$. This way, for any $\Theta = (\Omega, \{\Gamma^{(k)}\}_{k=0}^K) \in \mathbb{H}$, we have

$$[\Theta]_{\mathbb{M}} = \left([\Omega]_{\mathcal{S}_\Omega}, \left\{ [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\}_{k=0}^K \right) \quad \text{and} \quad [\Theta]_{\mathbb{M}^\perp} = \left([\Omega]_{\mathcal{S}_\Omega^c}, \left\{ [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}^c} \right\}_{k=0}^K \right).$$

Finally, we define our metric on the estimation error. For $\Theta = (\Omega, \{\Gamma^{(k)}\}_{k=0}^K) \in \mathbb{H}$, let

$$H(\Theta) := \sum_{k=0}^K \alpha_k \|\Omega + \Gamma^{(k)}\|_{\text{F}}^2. \quad (11)$$

We define

$$\hat{\Delta} := \hat{\Theta} - \Theta^\star = \left(\hat{\Omega} - \Omega^\star, \left\{ \hat{\Gamma}^{(k)} - \Gamma^{(k)\star} \right\}_{k=0}^K \right),$$

where $\hat{\Theta}$ is the Trans-Glasso estimator (2). Then our goal is to control

$$H(\hat{\Delta}) = \sum_{k=0}^K \alpha_k \left\| \hat{\Omega} - \Omega^\star + \hat{\Gamma}^{(k)} - \Gamma^{(k)\star} \right\|_{\text{F}}^2 = \sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2.$$

G.2 Useful lemmas

Now we collect several useful lemmas, whose proofs are deferred to the end of the section.

We begin by demonstrating that $\Phi(\cdot)$, defined in (4), is decomposable with respect to $(\mathbb{M}, \mathbb{M}^\perp)$.

Lemma 8. *We have*

$$\Phi(\Theta + \Theta') = \Phi(\Theta) + \Phi(\Theta') \quad \text{for any } \Theta \in \mathbb{M}, \Theta' \in \mathbb{M}^\perp.$$

The following lemma relates $\Phi(\Theta)$ with $H(\Theta)$ defined in (11) for any $\Theta \in \mathbb{M}$.

Lemma 9. *For any $\Theta \in \mathbb{M}$, we have*

$$\Phi(\Theta) \leq \sqrt{2} \sqrt{s + (K+1)h} \sqrt{H(\Theta)}.$$

For $\Delta_\Theta = (\Delta_\Omega, \{\Delta_{\Gamma^{(k)}}\}_{k=0}^K) \in \mathbb{H}$ such that $\Delta_\Omega + \Delta_{\Gamma^{(k)}} + \Omega^{(k)} > 0$ for all $0 \leq k \leq K$, we define

$$\mathcal{R}(\Delta_\Theta) := \mathcal{L}(\Theta^* + \Delta_\Theta) - \mathcal{L}(\Theta^*) - \langle \nabla \mathcal{L}(\Theta^*), \Delta_\Theta \rangle$$

to be the residual of $\mathcal{L}(\cdot)$ around Θ^* , where $\mathcal{L}(\cdot)$ is defined in (3). The following lemma claims that $\mathcal{R}(\cdot)$ is locally strongly convex with respect to the geometry defined by $H(\cdot)$.

Lemma 10. *Let $\Delta_\Theta = (\Delta_\Omega, \{\Delta_{\Gamma^{(k)}}\}_{k=0}^K) \in \mathbb{H}$ be such that $\Delta_\Omega + \Delta_{\Gamma^{(k)}} + \Omega^{(k)} > 0$ and assume that $\|\Delta_\Omega + \Delta_{\Gamma^{(k)}}\|_2 \leq M_{\text{op}} + M_\Omega$ for all $0 \leq k \leq K$. Then, we have*

$$\mathcal{R}(\Delta_\Theta) \geq \frac{\kappa}{2} H(\Delta_\Theta),$$

where $\kappa = (2M_\Omega + M_{\text{op}})^{-2}$.

Last but not least, define the event

$$\mathbb{G}(\lambda_M) := \left\{ \frac{\lambda_M}{2} \geq \Phi^*(\nabla \mathcal{L}(\Theta^*)) \right\}. \quad (12)$$

The following lemma asserts that this event happens with high probability.

Lemma 11. *When (17) holds and λ_M satisfies (18), we have that $\mathbb{P}\{\mathbb{G}(\lambda_M)\} \geq 1 - \delta$.*

G.3 Remaining proof

Now we are ready to prove Theorem 1. As Lemma 11 asserts that $\mathbb{G}(\lambda_M)$ holds with high probability, thus we only need to prove the conclusion under the assumption that $\mathbb{G}(\lambda_M)$ holds. Throughout the proof, we assume the event $\mathbb{G}(\lambda_M)$ holds.

For any $\Delta_\Theta = (\Delta_\Omega, \{\Delta_{\Gamma^{(k)}}\}_{k=0}^K) \in \mathbb{H}$, we define the objective difference

$$F(\Delta_\Theta) := \mathcal{L}(\Theta^\star + \Delta_\Theta) - \mathcal{L}(\Theta^\star) + \lambda_M \{\Phi(\Theta^\star + \Delta_\Theta) - \Phi(\Theta^\star)\}.$$

Note that by definition, we have $F(\hat{\Delta}_\Theta) \leq 0$. It suffices to show that for all Δ_Θ such that $\Delta_\Theta + \Theta^\star \in \mathcal{C}(M_{\text{op}})$ and $H(\Delta_\Theta) > \frac{18(s+(K+1)h)\lambda_M^2}{\kappa^2}$, we have $F(\Delta_\Theta) > 0$.

It is easy to see that any such Δ_Θ obeys

$$\Delta\Omega + \Delta\Gamma_k + \Omega^{(k)} = \Omega - \Omega^\star + \Gamma_k - \Gamma^{(k)\star} + \Omega^{(k)} = \Omega + \Gamma_k > 0,$$

and for all $0 \leq k \leq K$

$$\|\Delta\Omega + \Delta\Gamma_k\|_2 = \|\Omega - \Omega^\star + \Gamma_k - \Gamma^{(k)\star}\|_2 = \|\Omega + \Gamma_k - \Omega^{(k)}\|_2 \leq \|\Omega + \Gamma_k\|_2 + \|\Omega^{(k)}\|_2 \leq M_{\text{op}} + M_\Omega,$$

where the last inequality follows the definition of M_{op} and M_Ω . These two taken together allows us to invoke Lemma 10 to obtain

$$F(\Delta_\Theta) \geq \langle \nabla \mathcal{L}(\Theta^\star), \Delta_\Theta \rangle + \frac{\kappa}{2} H(\Delta_\Theta) + \lambda_M \{\Phi(\Theta^\star + \Delta_\Theta) - \Phi(\Theta^\star)\},$$

where $\kappa = (2M_\Omega + M_{\text{op}})^{-2}$.

In addition, combining Lemma 8 and the fact that $[\Theta^\star]_{\mathbb{M}^\perp} = 0$ with Wainwright (2019, Lemma 9.14), we have

$$\Phi(\Theta^\star + \Delta_\Theta) - \Phi(\Theta^\star) \geq \Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}}) \quad (13)$$

for any $\Delta_\Theta \in \mathbb{H}$. This way, we have obtained a lower bound for $\Phi(\Theta^\star + \Delta_\Theta) - \Phi(\Theta^\star)$.

Take the previous two displays together to reach

$$F(\Delta_\Theta) \geq \langle \nabla \mathcal{L}(\Theta^\star), \Delta_\Theta \rangle + \frac{\kappa}{2} H(\Delta_\Theta) + \lambda_M \{\Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}})\}.$$

Recall the definition of Φ^* in (8), under the assumption that $\mathbb{G}(\lambda_M)$ is true, we have

$$\begin{aligned}
F(\Delta_\Theta) &\geq \frac{\kappa}{2} H(\Delta_\Theta) - |\langle \nabla \mathcal{L}(\Theta^*), \Delta_\Theta \rangle| + \lambda_M \{\Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}})\} \\
&\geq \frac{\kappa}{2} H(\Delta_\Theta) - \Phi(\Delta_\Theta) \Phi^*(\nabla \mathcal{L}(\Theta^*)) + \lambda_M \{\Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}})\} \\
&\geq \frac{\kappa}{2} H(\Delta_\Theta) - \frac{\lambda_M}{2} \Phi(\Delta_\Theta) + \lambda_M \{\Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}})\} \\
&= \frac{\kappa}{2} H(\Delta_\Theta) - \frac{\lambda_M}{2} \{\Phi([\Delta_\Theta]_{\mathbb{M}}) + \Phi([\Delta_\Theta]_{\mathbb{M}^\perp})\} + \lambda_M \{\Phi([\Delta_\Theta]_{\mathbb{M}^\perp}) - \Phi([\Delta_\Theta]_{\mathbb{M}})\} \\
&= \frac{\kappa}{2} H(\Delta_\Theta) - \frac{\lambda_M}{2} \{3\Phi([\Delta_\Theta]_{\mathbb{M}}) - \Phi([\Delta_\Theta]_{\mathbb{M}^\perp})\} \\
&\geq \frac{\kappa}{2} H(\Delta_\Theta) - \frac{3\lambda_M}{2} \Phi([\Delta_\Theta]_{\mathbb{M}}). \tag{14}
\end{aligned}$$

By Lemma 9, we have

$$\Phi([\Delta_\Theta]_{\mathbb{M}}) \leq \sqrt{s + (K+1)h} \sqrt{H([\Delta_\Theta]_{\mathbb{M}})} \leq \sqrt{2} \sqrt{s + (K+1)h} \sqrt{H(\Delta_\Theta)},$$

where the last inequality follows from $H(\Delta_\Theta) = H([\Delta_\Theta]_{\mathbb{M}}) + H([\Delta_\Theta]_{\mathbb{M}^\perp}) \geq H([\Delta_\Theta]_{\mathbb{M}})$.

Plugging the above inequality into (14), we arrive at the conclusion that

$$\begin{aligned}
F(\Delta_\Theta) &\geq \frac{\kappa}{2} H(\Delta_\Theta) - \frac{3\lambda_M}{\sqrt{2}} \sqrt{s + (K+1)h} \sqrt{H(\Delta_\Theta)} \\
&= \frac{1}{2} \sqrt{H(\Delta_\Theta)} \left\{ \kappa \sqrt{H(\Delta_\Theta)} - 3\sqrt{2}\lambda_M \sqrt{s + (K+1)h} \right\} > 0,
\end{aligned}$$

where the last relation arises from the assumption

$$H(\Delta_\Theta) > \frac{18(s + (K+1)h)\lambda_M^2}{\kappa^2}. \tag{15}$$

This finishes the proof.

G.4 Proof of Useful Lemmas

In this section, we collect the proof of useful lemmas.

G.4.1 Proof of Lemma 8

By the definition of $\Phi(\cdot)$ and the fact that $\Theta \in \mathbb{M}$ and $\Theta' \in \mathbb{M}^\perp$, we have

$$\begin{aligned}
\Phi(\Theta + \Theta') &= \Phi([\Theta]_{\mathbb{M}} + [\Theta']_{\mathbb{M}^\perp}) \\
&= \left| [\Omega]_{\mathcal{S}_\Omega} + [\Omega']_{\mathcal{S}_\Omega^c} \right|_1 + \sum_{k=0}^K \sqrt{\alpha_k} \left| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} + [\Gamma^{(k)'}]_{\mathcal{S}_{\Gamma^{(k)}}^c} \right|_1 \\
&= \left| [\Omega]_{\mathcal{S}_\Omega} \right|_1 + \left| [\Omega']_{\mathcal{S}_\Omega^c} \right|_1 + \sum_{k=0}^K \sqrt{\alpha_k} \left| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right|_1 + \sum_{k=0}^K \sqrt{\alpha_k} \left| [\Gamma^{(k)'}]_{\mathcal{S}_{\Gamma^{(k)}}^c} \right|_1 \\
&= \Phi([\Theta]_{\mathbb{M}}) + \Phi([\Theta']_{\mathbb{M}^\perp}) \\
&= \Phi(\Theta) + \Phi(\Theta').
\end{aligned}$$

G.4.2 Proof of Lemma 9

For any $\Theta \in \mathbb{M}$, we have $\Omega = [\Omega]_{\mathcal{S}_\Omega}$ and $\Gamma^{(k)} = [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}}$. Thus

$$\begin{aligned}
\Phi(\Theta) &= \left| [\Omega]_{\mathcal{S}_\Omega} \right|_1 + \sum_{k=0}^K \sqrt{\alpha_k} \left| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right|_1 \\
&\leq |\mathcal{S}_\Omega|^{\frac{1}{2}} \left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F + \sum_{k=0}^K \sqrt{\alpha_k} |\mathcal{S}_{\Gamma^{(k)}}|^{\frac{1}{2}} \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F \quad (\text{Jensen's inequality}) \\
&\stackrel{(ii)}{\leq} \sqrt{s} \left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F + \sqrt{h} \sum_{k=0}^K \sqrt{\alpha_k} \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F. \quad (\text{by Assumption 1})
\end{aligned}$$

Furthermore, by Jensen's inequality, we have

$$\begin{aligned}
\frac{1}{2} \Phi^2(\Theta) &\leq s \left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F^2 + h \left(\sum_{k=0}^K \sqrt{\alpha_k} \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F \right)^2 \\
&\leq s \left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F^2 + h(K+1) \sum_{k=0}^K \alpha_k \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F^2 \\
&\leq (s + (K+1)h) \left\{ \left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F^2 + \sum_{k=0}^K \alpha_k \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F^2 \right\} \\
&= (s + (K+1)h) \sum_{k=0}^K \alpha_k \left(\left\| [\Omega]_{\mathcal{S}_\Omega} \right\|_F^2 + \left\| [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_F^2 \right).
\end{aligned}$$

By the assumption that the supports of $\Omega^\star$ and $\Gamma^{(k)\star}$ are disjoint and the fact that $\Omega = [\Omega]_{\mathcal{S}_\Omega}$ and $\Gamma^{(k)} = [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}}$, we then have

$$\begin{aligned}\Phi^2(\Theta) &\leq 2(s + (K + 1)h) \sum_{k=0}^K \alpha_k \left\| [\Omega]_{\mathcal{S}_\Omega} + [\Gamma^{(k)}]_{\mathcal{S}_{\Gamma^{(k)}}} \right\|_{\mathbb{F}}^2 \\ &\leq 2(s + (K + 1)h) \sum_{k=0}^K \alpha_k \left\| \Omega + \Gamma^{(k)} \right\|_{\mathbb{F}}^2 \\ &= 2(s + (K + 1)h) H(\Theta).\end{aligned}$$

G.4.3 Proof of Lemma 10

By Lemma 5, we have

$$\begin{aligned}\mathcal{R}(\Delta_\Theta) &= \sum_{k=0}^K \frac{\alpha_k}{2} \text{vec}(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \\ &\quad \times \left\{ \left(\Omega^{(k)} + t_k(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right)^{-1} \otimes \left(\Omega^{(k)} + t_k(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right)^{-1} \right\} \\ &\quad \times \text{vec}(\Delta_\Omega + \Delta_{\Gamma^{(k)}}), \quad (16)\end{aligned}$$

where $t_k \in (0, 1)$, $0 \leq k \leq K$. Since $\gamma_{\min}(A^{-1} \otimes A^{-1}) = \|A\|_2^{-2}$ for any $A > 0$, we have

$$\begin{aligned}\gamma_{\min} &\left(\left(\Omega^{(k)} + t_k(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right)^{-1} \otimes \left(\Omega^{(k)} + t_k(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right)^{-1} \right) \\ &= \left\{ \left\| \Omega^{(k)} + t_k(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right\|_2 \right\}^{-2} \\ &\geq \left\{ \left\| \Omega^{(k)} \right\|_2 + t_k \left\| \Delta_\Omega + \Delta_{\Gamma^{(k)}} \right\|_2 \right\}^{-2} \\ &\geq \left\{ \left\| \Omega^{(k)} \right\|_2 + \left\| \Delta_\Omega + \Delta_{\Gamma^{(k)}} \right\|_2 \right\}^{-2} \\ &\geq (2M_\Omega + M_{\text{op}})^{-2},\end{aligned}$$

where the last line follows the definition of M_Ω in (16) and the assumption that $\left\| \Delta_\Omega + \Delta_{\Gamma^{(k)}} \right\|_2 \leq M_\Omega + M_{\text{op}}$ for all $0 \leq k \leq K$. Let $\kappa = (2M_\Omega + M_{\text{op}})^{-2}$. Then

$$\mathcal{R}(\Delta_\Theta) \geq \frac{\kappa}{2} \sum_{k=0}^K \alpha_k \left\| \text{vec}(\Delta_\Omega + \Delta_{\Gamma^{(k)}}) \right\|_2^2.$$

The final result follows by noting that $\|\text{vec}(\Delta_\Omega + \Delta_{\Gamma^{(k)}})\|_2 = \|\Delta_\Omega + \Delta_{\Gamma^{(k)}}\|_F$.

G.4.4 Proof of Lemma 11

We first state and prove the following lemma that gives the closed-form expression of $\Phi^*(\cdot)$.

Lemma 12. *For the dual norm defined in (8), we have*

$$\Phi^*(\Theta) = \max \left\{ |\Omega|_\infty, \max_{0 \leq k \leq K} \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \right\}.$$

Proof. For any $\Theta, \Theta' \in \mathbb{H}$, we have

$$\begin{aligned} \langle \Theta, \Theta' \rangle &= \langle \Omega, \Omega' \rangle + \sum_{k=0}^K \langle \Gamma^{(k)}, \Gamma^{(k)'} \rangle \\ &\leq |\Omega|_\infty |\Omega'|_1 + \sum_{k=0}^K |\Gamma^{(k)}|_\infty |\Gamma'|_1 \\ &= |\Omega|_\infty |\Omega'|_1 + \sum_{k=0}^K \sqrt{\alpha_k} |\Gamma'|_1 \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \\ &\leq |\Omega|_\infty |\Omega'|_1 + \left\{ \max_{0 \leq k \leq K} \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \right\} \sum_{k=0}^K \sqrt{\alpha_k} |\Gamma'|_1 \\ &\leq \max \left\{ |\Omega|_\infty, \max_{0 \leq k \leq K} \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \right\} \left(|\Omega'|_1 + \sum_{k=0}^K \sqrt{\alpha_k} |\Gamma'|_1 \right) \\ &= \max \left\{ |\Omega|_\infty, \max_{0 \leq k \leq K} \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \right\} \Phi(\Theta'). \end{aligned}$$

Thus, we have

$$\Phi^*(\Theta) := \sup_{\Theta': \Phi(\Theta') \leq 1} \langle \Theta, \Theta' \rangle \leq \max \left\{ |\Omega|_\infty, \max_{0 \leq k \leq K} \frac{|\Gamma^{(k)}|_\infty}{\sqrt{\alpha_k}} \right\}.$$

Finally, it is easy to see that the equality is achievable. □

Now we are ready to prove Lemma 11. Note that

$$\begin{aligned} \nabla_\Omega \mathcal{L}(\Theta) &= \sum_{k=0}^K \alpha_k \left(\hat{\Sigma}^{(k)} - (\Omega + \Gamma^{(k)})^{-1} \right), \\ \nabla_{\Gamma^{(k)}} \mathcal{L}(\Theta) &= \alpha_k \left(\hat{\Sigma}^{(k)} - (\Omega + \Gamma^{(k)})^{-1} \right) \quad \text{for all } 0 \leq k \leq K. \end{aligned}$$

Then by Lemma 12, we have

$$\mathbb{G}(\lambda_M) = \left\{ \frac{\lambda_M}{2} \geq \max \left\{ \left| \sum_{k=0}^K \alpha_k \left(\hat{\Sigma}^{(k)} - \Sigma^{(k)} \right) \right|_{\infty}, \max_{0 \leq k \leq K} \sqrt{\alpha_k} \left| \hat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_{\infty} \right\} \right\}. \quad (17)$$

By Lemma 3 and the union bound, we have

$$\left| \sum_{k=0}^K \alpha_k \left(\hat{\Sigma}^{(k)} - \Sigma^{(k)} \right) \right|_{\infty} \leq 80M_{\Sigma} \max \left\{ \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2N}}, \frac{\log(2(K+2)d^2/\delta)}{N} \right\}$$

and

$$\left| \hat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_{\infty} \leq 80M_{\Sigma} \max \left\{ \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2n_k}}, \frac{\log(2(K+2)d^2/\delta)}{n_k} \right\}$$

for all $0 \leq k \leq K$ hold simultaneously with probability at least $1 - \delta$. When $\min_{0 \leq k \leq K} n_k$

is large enough such that

$$\frac{\log(2(K+2)d^2/\delta)}{\min_{0 \leq k \leq K} n_k} \leq \frac{1}{2},$$

we then have that

$$\begin{aligned} \left| \sum_{k=0}^K \alpha_k \left(\hat{\Sigma}^{(k)} - \Sigma^{(k)} \right) \right|_{\infty} &\leq 80M_{\Sigma} \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2N}}, \\ \left| \hat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_{\infty} &\leq 80M_{\Sigma} \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2n_k}} \quad \text{for all } 0 \leq k \leq K. \end{aligned} \quad (18)$$

hold simultaneously with probability at least $1 - \delta$. Note that (18) implies that

$$\sqrt{\alpha_k} \left| \hat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_{\infty} \leq 80M_{\Sigma} \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2N}} \quad \text{for all } 0 \leq k \leq K.$$

Thus, when

$$\lambda_M \geq 160M_{\Sigma} \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2N}}, \quad (19)$$

by (17), we have $\mathbb{G}(\lambda_M)$ hold with probability $1 - \delta$.

H Corollary 1 and Its Proof

The following corollary of Theorem 10 in Zhao et al. (2022) provides a high-probability error bound for the D-Trace Loss estimator.

Corollary 1. *Suppose that Assumption 1 and Assumption 2 hold. Let $C_\gamma = M_\Omega^{-1}$. Furthermore, suppose that for a given $\delta \in (0, 1]$ and all $k \in [K]$ we have*

$$\frac{\log(2(K+1)d^2/\delta)}{\min\{n_k, n_0\}} \leq \min \left\{ 1, \frac{\gamma_{\min}^2(\Sigma^{(k)})\gamma_{\min}^2(\Sigma^{(0)})}{C'h^2M_\Sigma^4} \right\}. \quad (20)$$

If

$$\lambda_\Gamma^{(k)} \geq 2C_\gamma [(C_\gamma + 2)M_\Sigma M_\Gamma + 2] M_\Sigma \sqrt{\frac{\log(2(K+1)d^2/\delta)}{2\min\{n_k, n_0\}}}$$

for all $k \in [K]$, then

$$\|\hat{\Psi}^{(k)} - \Psi^{(k)}\|_F \leq \frac{C''\sqrt{h}\lambda_\Gamma^{(k)}}{\gamma_{\min}(\Sigma^{(k)})\gamma_{\min}(\Sigma^{(0)})} \quad \text{for all } 0 \leq k \leq K$$

hold simultaneously with probability at least $1 - \delta$, where C', C'' are universal constants that depend on C_γ .

Proof. The D-Trace Loss estimator is a special case of the FuDGE estimator proposed in Zhao et al. (2019, 2022). We adapt the proof of Theorem 10 in Zhao et al. (2022) here.

Given $\delta \in (0, 1]$, we define the event

$$\mathcal{E}^1(\delta) := \left\{ \left| \hat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_\infty \leq C_\gamma M_\Sigma \cdot \max \left\{ \sqrt{\frac{\log(2(K+1)d^2/\delta)}{2n_k}}, \frac{\log(2(K+1)d^2/\delta)}{n_k} \right\} \right. \\ \left. \text{for all } 0 \leq k \leq K \right\},$$

where C_γ is a universal constant. By Lemma 4 and the union bound, we have $\mathbb{P}(\mathcal{E}^1(\delta)) \geq 1 - \delta$. In the following, we work on the event $\mathcal{E}^1(\delta)$.

Let

$$\psi_k = C_\gamma M_\Sigma \sqrt{\frac{\log(2(K+1)d^2/\delta)}{2 \min\{n_k, n_0\}}}.$$

On the event $\mathcal{E}^1$ and when the condition (20) holds with $C' = 4096C_\gamma^2(C_\gamma + 2)^2$, we have

$$\max \left\{ \left| \widehat{\Sigma}^{(k)} - \Sigma^{(k)} \right|_\infty, \left| \widehat{\Sigma}^{(0)} - \Sigma^{(0)} \right|_\infty \right\} \leq \psi_k.$$

When (20) holds, we have

$$\psi_k \leq C_\gamma \cdot \max_{0 \leq k \leq K} \left| \Sigma^{(k)} \right|_\infty \quad \text{and} \quad \psi_k \leq \frac{\gamma_{\min}(\Sigma^{(k)})\gamma_{\min}(\Sigma^{(0)})}{64(C_\gamma + 2)hM_\Sigma}.$$

Thus, we have

$$\kappa_{\mathcal{L}}^{(k)} := \frac{1}{2}\gamma_{\min}(\Sigma^{(k)})\gamma_{\min}(\Sigma^{(0)}) - 16h(\psi_k^2 + 2\psi_k M_\Sigma) \geq \frac{1}{4}\gamma_{\min}(\Sigma^{(k)})\gamma_{\min}(\Sigma^{(0)}),$$

where $\kappa_{\mathcal{L}}^{(k)}$ is the restricted convexity parameter defined in the proof of Theorem 10 of [Zhao et al. \(2022\)](#). Set

$$\lambda_\Gamma^{(k)} \geq 4[(C_\gamma + 2)M_\Sigma M_\Gamma + 1]\psi_k.$$

Following the proof of Theorem 10 of [Zhao et al. \(2022\)](#), we have

$$\left\| \widehat{\Psi}^{(k)} - \Psi^{(k)} \right\|_F^2 \leq \frac{18h}{\frac{1}{16}\gamma_{\min}^2(\Sigma^{(k)})\gamma_{\min}^2(\Sigma^{(0)})} \left(\lambda_\Gamma^{(k)} \right)^2,$$

which completes the proof. □

I Proof of Lemma 2

Let $R = \lfloor s/d \rfloor$ and let $r = s - Rd$. We follow the construction of the hard-case collection of precision matrices in Step 2 of Proof of Theorem 4.1 in [Cai et al. \(2016\)](#) to get $\mathcal{F}_\star$. We set $M_{n,p}$ and $c_{n,p}$ therein as $M_{n,p} = \text{some constant}$ and $c_{n,p} = R$. By (4.13) and (4.15) in [Cai](#)

et al. (2016), for any $\Omega \in \mathcal{F}_\star$, we have

$$0 < c_1 \leq \gamma_{\min}(\Omega) \leq \gamma_{\max}(\Omega) \leq c_2 < \infty.$$

Furthermore, we have $|\Omega|_0 \leq c_{n,p}d \leq s$. Thus, $\mathcal{F}_\star \subseteq \mathcal{G}_1$. Following Theorem 6.1 of Cai et al. (2016), we have

$$\inf_{\hat{\Omega}} \sup_{\Omega \in \mathcal{G}_1} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega \right\|_{\text{F}}^2 \right] \geq \inf_{\hat{\Omega}} \sup_{\Omega \in \mathcal{F}_\star} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega \right\|_{\text{F}}^2 \right] \gtrsim dc_{n,p} \frac{\log d}{n} = \frac{dR}{s} \cdot \frac{s \log d}{n} \geq \frac{1}{2} \frac{s \log d}{n},$$

which completes the proof.

J Proof of Lemma 3

Without loss of generality, we assume that d and h are even positive numbers. Let $m = d/2$.

By Lemma 6, there exist $\{b^0, b^1, \dots, b^M\} \subseteq \mathbb{R}^{m^2}$ such that

- (i) $\|b^j\|_0 \leq \frac{h}{2}$ and $\|b^j\|_2 \leq 1$ for all $0 \leq j \leq M$;
- (ii) $b^0 = 0$;
- (iii) $\|b^j - b^k\|_2 \geq \frac{1}{4}$ for all $0 \leq j \neq k \leq M$;
- (iv) $\log(M+1) \geq \frac{h}{4} \log \left(\frac{d^2/4 - h/2}{h/2} \right)$.

For $j = 0, \dots, M$, let $B^j \in \mathbb{R}^{m \times m}$ be such that $\text{vec}(B^j) = \delta \cdot b^j$, where δ is a positive number that depends on n, d, h and will be specified later. By (23), $d \geq 4h$ and

$$\log(M+1) \geq \frac{h}{4} \log \left(\frac{d^2/4 - h/2}{h/2} \right) = \frac{h}{4} \log \left(\frac{d^2}{2h} - 1 \right) \geq \frac{h}{4} \log \left(\frac{d^2}{4h} \right) \geq \frac{h}{4} \log d.$$

Furthermore, we have

$$\begin{aligned} |B^j|_0 &\leq \frac{h}{2}, \quad \|B^j\|_{\text{F}} \leq \delta, \quad \text{for } 0 \leq j \leq M, \\ B^0 &= 0, \quad \|B^j - B^k\|_{\text{F}} \geq \frac{\delta}{4}, \quad \text{for } 0 \leq j \neq k \leq M. \end{aligned}$$

For $j = 0, \dots, M$, let

$$\Omega^j = \begin{bmatrix} I_m & B^j \\ (B^j)^\top & I_m \end{bmatrix}.$$

We next verify that $\Omega^j \in \mathcal{G}_2$ when

$$\delta \leq \min \left\{ \frac{C_\Gamma}{\sqrt{2h}}, \frac{1-c_1}{2}, \frac{c_2-1}{2} \right\}, \quad (21)$$

where $\mathcal{G}_2$ is defined in (22). First note that $\text{diag}(\Omega^j) = I_d$, $|\Omega^j - I_d|_0 = 2|B^j|_0 \leq h$, and

$$|\Omega^j - I_d|_1 = 2|B^j|_1 \leq 2\sqrt{\frac{h}{2}} \|B^j\|_F \leq \sqrt{2h}\delta \leq C_\Gamma,$$

using the choice of δ in (21). Furthermore, we have

$$\begin{aligned} \gamma_{\min}(\Omega^j) &= \gamma_{\min} \left(\begin{bmatrix} I_m & 0 \\ 0 & I_m \end{bmatrix} + \begin{bmatrix} 0 & B^j \\ (B^j)^\top & 0 \end{bmatrix} \right) \\ &\geq 1 - \left\| \begin{bmatrix} 0 & B^j \\ (B^j)^\top & 0 \end{bmatrix} \right\|_2 \\ &\geq 1 - \left\| \begin{bmatrix} 0 & B^j \\ (B^j)^\top & 0 \end{bmatrix} \right\|_F \\ &= 1 - 2\|B^j\|_F \\ &\geq c_1, \end{aligned}$$

again using the choice of δ in (21). Similarly, we have $\gamma_{\max}(\Omega^j) \leq c_2$. Therefore, when (21)

holds, we have that $\Omega^j \in \mathcal{G}_2$.

Let $\mathbb{P}_j$ denote the probability measure of $N(0, \{\Omega^j\}^{-1})$ and let $\mathbb{P}_j^n$ denote the product probability measure of $(X_1, \dots, X_n)$ where $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \mathbb{P}_j$. Note that $\mathbb{P}_0 = N(0, I_d)$.

When $\delta \leq 1/4$, Lemma 7 gives us

$$D_{\text{KL}}(\mathbb{P}_j^n \parallel \mathbb{P}_0^n) = n D_{\text{KL}}(\mathbb{P}_j \parallel \mathbb{P}_0) \leq \frac{16}{15} n \|B^j\|_{\text{F}}^2 \leq \frac{16}{15} n \delta^2,$$

which implies that

$$\frac{1}{M} \sum_{j=1}^M D_{\text{KL}}(\mathbb{P}_j^n \parallel \mathbb{P}_0^n) \leq \frac{16}{15} n \delta^2. \quad (22)$$

From condition (23), we have $h \log d \geq 8 \log 3$ and thus $\log(M+1) \geq \log 9$, which implies that $M \geq 8$. Furthermore, we have

$$\log M = \frac{\log M}{\log(M+1)} \log(M+1) \geq \frac{\log 8}{\log 9} \log(M+1) \geq \frac{4}{5} \log(M+1) \geq \frac{h}{5} \log d.$$

We set

$$\delta = \frac{1}{4\sqrt{2}} \sqrt{\frac{h \log d}{n}}.$$

By (23), we have that (21) holds and that $\delta \leq 1/4$. Thus (22) holds, implying

$$\frac{1}{M} \sum_{j=1}^M D_{\text{KL}}(\mathbb{P}_j^n \parallel \mathbb{P}_0^n) \leq \frac{1}{30} h \log d \leq \frac{1}{6} \log M. \quad (23)$$

In addition, we have

$$\|B^j - B^k\|_{\text{F}} \geq 2 \cdot \frac{1}{32\sqrt{2}} \sqrt{\frac{h \log d}{n}} \quad \text{for all } 0 \leq j \neq k \leq M. \quad (24)$$

Note that

$$\frac{\log(M+1) - \log 2}{\log M} - \frac{1}{6} \geq \frac{\log M - \log 2}{\log M} - \frac{1}{6} \geq \frac{\log 8 - \log 2}{\log 8} - \frac{1}{6} = \frac{1}{2}. \quad (25)$$

Following Section 2.2 and Corollary 2.6 in [Tsybakov \(2008\)](#), combined with (23)–(25), we have

$$\min_{\hat{\Omega}} \sup_{\Omega \in \mathcal{G}_2} \mathbb{E} \left[\|\hat{\Omega} - \Omega\|_{\text{F}}^2 \right] \geq \frac{1}{2} \left(\frac{1}{32\sqrt{2}} \sqrt{\frac{h \log d}{n}} \right)^2 = \frac{1}{4096} \cdot \frac{h \log d}{n},$$

which completes the proof.

K Proof of Theorem 2

The upper bound follows directly from Corollary 2. To prove the lower bound, we assume that $\mathcal{P}_k = N\left(0, (\Omega^{(k)})^{-1}\right)$.

First, when $h = 0$, we have N i.i.d. samples from $\Omega^{(0)}$ where

$$\Omega^{(0)} \in \{\Omega \in \mathbb{R}^{d \times d} : \Omega \succ 0, |\Omega|_0 \leq s, \|\Omega\|_2 \leq M_\Omega, |\Omega^{-1}|_\infty \leq M_\Sigma\} := \tilde{\mathcal{G}}'$$

for some positive universal constants M_Ω and M_Σ . Let

$$\mathcal{G}' = \left\{ \Omega \in \mathbb{R}^{d \times d} : \Omega \succ 0, |\Omega|_0 \leq s, 0 < \frac{1}{M_\Sigma} \leq \gamma_{\min}(\Omega) \leq \gamma_{\max}(\Omega) \leq M_\Omega < \infty \right\}.$$

Since $|\Omega^{-1}|_\infty \leq \|\Omega^{-1}\|_2 = \{\gamma_{\min}(\Omega)\}^{-1} \leq M_\Sigma$ for any $\Omega \in \mathcal{G}'$, we have $\mathcal{G}' \subseteq \tilde{\mathcal{G}}'$. Thus, when $s \geq d \geq c'N^\beta$, for some universal constants $c' > 0$ and $\beta > 1$ and

$$\lceil s/d \rceil = o\left(\frac{N}{(\log d)^{\frac{3}{2}}}\right)$$

it follows by Lemma 2 that

$$\inf_{\hat{\Omega}} \sup_{\Omega^{(0)} \in \tilde{\mathcal{G}}'} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega^{(0)} \right\|_F^2 \right] \geq \inf_{\hat{\Omega}} \sup_{\Omega^{(0)} \in \mathcal{G}'} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega^{(0)} \right\|_F^2 \right] \gtrsim \frac{s \log d}{N}. \quad (26)$$

If $\Omega^{(k)} = \Omega^\star = I_d$ and $\Gamma^{(k)\star} = 0$ for every $k \in [K]$, then samples from source distributions cannot be used to estimate $\Omega^{(0)}$. Therefore, we must depend solely on samples from the target distribution to estimate $\Omega^{(0)}$. Note that now we have

$$\Omega^{(0)} \in \{\Omega \in \mathbb{S}^{d \times d} : \Omega \succ 0, \Omega = I_d + \Delta, \Delta_{jj} = 0 \text{ for all } 1 \leq j \leq d,$$

$$|\Delta|_0 \leq h, |\Delta|_1 \leq M_\Gamma, \|\Omega\|_2 \leq M_\Omega, |\Omega^{-1}|_\infty \leq M_\Sigma\} := \tilde{\mathcal{G}}''.$$

Let

$$\mathcal{G}'' = \left\{ \Omega \in \mathbb{S}^{d \times d} : \Omega \succ 0, \Omega = I_d + \Delta, \Delta_{jj} = 0 \text{ for all } 1 \leq j \leq d, \right. \\ \left. |\Delta|_0 \leq h, |\Delta|_1 \leq M_\Gamma, 0 < \frac{1}{M_\Sigma} \leq \gamma_{\min}(\Omega) \leq \gamma_{\max}(\Omega) \leq M_\Omega < \infty \right\}.$$

Given that for any $\Omega \in \mathcal{G}''$, it holds that $|\Omega^{-1}|_\infty \leq \|\Omega^{-1}\|_2 = \{\gamma_{\min}(\Omega)\}^{-1} \leq M_\Sigma$, we can deduce that $\mathcal{G}'' \subseteq \tilde{\mathcal{G}}''$. By Lemma 3, when $M_\Omega, M_\Sigma > 1$ and

$$d \geq 4h, h \log d \geq 8 \log 3, \frac{h \log d}{n_0} \leq \min \left\{ 2, 8 \left(1 - \frac{1}{M_\Sigma} \right)^2, 8(1 - M_\Omega)^2 \right\}, h \sqrt{\frac{\log d}{n_0}} \leq 4M_\Gamma,$$

we have

$$\inf_{\hat{\Omega}} \sup_{\Omega^{(0)} \in \tilde{\mathcal{G}}''} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega^{(0)} \right\|_F^2 \right] \geq \inf_{\hat{\Omega}} \sup_{\Omega^{(0)} \in \mathcal{G}''} \mathbb{E} \left[\left\| \hat{\Omega} - \Omega^{(0)} \right\|_F^2 \right] \gtrsim \frac{h \log d}{n_0}. \quad (27)$$

The final result follows from (26) and (27).

L Additional Optimization Details

In this section, we provide more details about the numerical algorithms introduced in Section 4.1.

We first discuss how to compute the updating steps (8) and (9). Note that (8) is equivalent to

$$\Omega_k^{(t)} = \arg \min_{\Omega_k \succ 0} \left\{ f_k(\Omega_k) + \rho \left\langle Z_k^{(t-1)}, \Omega_k - \Omega^{(t-1)} - \Gamma_k^{(t-1)} \right\rangle + \frac{\rho}{2} \left\| \Omega_k - \Omega^{(t-1)} - \Gamma_k^{(t-1)} \right\|_F^2 \right\} \\ = \arg \min_{\Omega_k \succ 0} \left\{ -\alpha_k \log \det(\Omega_k) + \frac{\rho}{2} \left\| \Omega_k - \tilde{C}_k^{(t-1)} \right\|_F^2 \right\}, \quad (28)$$

where $\tilde{C}_k^{(t-1)} = -Z_k^{(t-1)} + \left(\Omega^{(t-1)} + \Gamma_k^{(t-1)} \right) - (\alpha_k/\rho) \hat{\Sigma}^{(k)}$ for $0 \leq k \leq K$. Taking the gradient with respect to Ω_k in (28) and setting it to zero gives

$$-\alpha_k \Omega_k^{-1} + \rho \Omega_k - \rho \tilde{C}_k^{(t-1)} = 0. \quad (29)$$

The matrix $\Omega_k^{(t)}$ is obtained by finding $\Omega_k \succ 0$ that satisfies (29). Let $\rho \tilde{C}_k^{(t-1)} = U \Lambda U^\top$,

$\Lambda = \text{diag}(\{\lambda_i\}_{i=1}^d)$, be the eigenvalue decomposition of $\rho\tilde{C}_k^{(t-1)}$. Following (Boyd et al., 2011, Section 6.5), we have

$$\Omega_k^{(t)} = U \text{diag} \left(\left\{ \frac{\lambda_i + \sqrt{\lambda_i^2 + 4\rho\alpha_k}}{2\rho} \right\}_{i=1}^d \right) U^\top.$$

On the other hand, computing (9) is equivalent to solving

$$\begin{aligned} \Omega^{(t)}, \{\Gamma_k^{(t)}\} &\in \arg \min_{\Omega, \{\Gamma^{(k)}\}} \left\{ \rho \sum_{k=0}^K \left\langle Z_k^{(t-1)}, \Omega_k^{(t)} - \Omega - \Gamma^{(k)} \right\rangle + \frac{\rho}{2} \sum_{k=0}^K \left\| \Omega_k^{(t)} - \Omega - \Gamma^{(k)} \right\|_F^2 \right. \\ &= \arg \min_{\Omega, \{\Gamma^{(k)}\}} \left\{ \frac{\rho}{2} \sum_{k=0}^K \left\| \Omega + \Gamma^{(k)} - \check{C}_k^{(t)} \right\|_F^2 + \lambda_M |\Omega|_1 + \lambda_M \sum_{k=0}^K \sqrt{\alpha_k} |\Gamma^{(k)}|_1 \right\} \end{aligned}$$

where $\check{C}_k^{(t)} = \Omega_k^{(t)} + Z_k^{(t-1)}$. Given $c = (c_1, \dots, c_K)$, let $\mathcal{S}(c)$ be defined as $\mathcal{S}(c) = (x^*, y^*)$

where

$$(x^*, y^*) \in \arg \min_{(x, y)} \left\{ \frac{\rho}{2} \sum_{k=0}^K (x + y_k - c_k)^2 + \lambda_M |x| + \lambda_M \sum_{k=0}^K \sqrt{\alpha_k} |y_k| \right\} \quad (30)$$

with $y = (y_1, \dots, y_K)$. With $\Omega^{(t)} = (\Omega_{jl}^{(t)})_{1 \leq j, l \leq d}$, $\Gamma_k^{(t)} = (\Gamma_{k,jl}^{(t)})_{1 \leq j, l \leq d}$, and $\check{C}_k^{(t)} = (\check{C}_{k,jl}^{(t)})_{1 \leq j, l \leq d}$,

we have

$$\Omega_{jl}^{(t)}, \{\Gamma_{k,jl}^{(t)}\}_{k=1}^K = \mathcal{S} \left(\left\{ \check{C}_{k,jl}^{(t)} \right\}_{k=1}^K \right), \quad \text{for } 1 \leq j, l \leq d.$$

To solve (30), we iteratively update x or y while fixing the other until convergence. For

$c \in \mathbb{R}$ and $\lambda \geq 0$, let

$$\text{ST}_\lambda(c) = \begin{cases} c - \lambda & \text{if } c > \lambda, \\ 0 & \text{if } |c| \leq \lambda, \\ c + \lambda & \text{if } c < -\lambda \end{cases} \quad (31)$$

be the soft-thresholding function. After initializing $x^{(0)}, y^{(0)}$, we repeat the following process until convergence:

$$x^{(r)} = \arg \min_x \left\{ \frac{\rho}{2} \sum_{k=0}^K \left(x + y_k^{(r-1)} - c_k \right)^2 + \lambda_M |x| + \lambda_M \sum_{k=0}^K \sqrt{\alpha_k} |y_k^{(r-1)}| \right\}$$

$$= \text{ST}_{\lambda_{\text{M}}/(\rho(K+1))} \left(\frac{1}{K+1} \sum_{k=0}^K (c_k - y_k^{(r-1)}) \right),$$

and

$$y^{(r)} = \arg \min_y \left\{ \frac{\rho}{2} \sum_{k=0}^K (x^{(r)} + y_k - c_k)^2 + \lambda_{\text{M}} |x^{(r)}| + \lambda_{\text{M}} \sum_{k=0}^K \sqrt{\alpha_k} |y_k| \right\}$$

$$\Leftrightarrow y_k^{(r)} = \text{ST}_{\lambda_{\text{M}}\sqrt{\alpha_k}/\rho} (c_k - x^{(r)}) \quad \text{for } 0 \leq k \leq K.$$

We then discuss the stopping criterion for the ADMM algorithm to solve Trans-MT-Glasso objective (2).

Stopping criterion for ADMM. Following (Boyd et al., 2011, Section 3.3.1), let $\epsilon^{\text{abs}} > 0$ be an absolute tolerance and $\epsilon^{\text{rel}} > 0$ be a relative tolerance, we then define the feasibility tolerance for primal feasibility condition $\epsilon^{\text{pri}} > 0$ and the feasibility tolerance for dual feasibility condition $\epsilon^{\text{dual}} > 0$ at iteration t as

$$\epsilon^{\text{pri}} = \epsilon^{\text{abs}} d\sqrt{K+1} + \epsilon^{\text{rel}} \max \left\{ \left(\sum_{k=0}^K \|\Omega_k^{(t)}\|_{\text{F}}^2 \right)^{\frac{1}{2}}, \left(\sum_{k=0}^K \|\Omega^{(t)} + \Gamma_k^{(t)}\|_{\text{F}}^2 \right)^{\frac{1}{2}} \right\},$$

$$\epsilon^{\text{dual}} = \epsilon^{\text{abs}} d\sqrt{K+1} + \epsilon^{\text{rel}} \left(\sum_{k=0}^K \|Z_k^{(t)}\|_{\text{F}}^2 \right)^{\frac{1}{2}}.$$

Besides, let

$$r^{\text{pri}} = \left(\sum_{k=0}^K \left\| \Omega_k^{(t)} - (\Omega^{(t)} + \Gamma_k^{(t)}) \right\|_{\text{F}}^2 \right)^{\frac{1}{2}} \quad \text{and}$$

$$r^{\text{dual}} = \rho \left(\sum_{k=0}^K \left\| (\Omega^{(t)} + \Gamma_k^{(t)}) - (\Omega^{(t-1)} + \Gamma_k^{(t-1)}) \right\|_{\text{F}}^2 \right)^{\frac{1}{2}}$$

be the primal and dual residuals at iteration t . We then stop the iteration if

$$r^{\text{pri}} \leq \epsilon^{\text{pri}} \quad \text{and} \quad r^{\text{dual}} \leq \epsilon^{\text{dual}}.$$

Stopping criterion for sub problem (30). The optimality conditions of problem (30) are

$$0 \in \rho \sum_{k=0}^K (x^* + y_k^* - c_k) + \lambda_M \partial |x^*|, \quad (32)$$

$$0 \in \rho (x^* + y^* - c_k) + \lambda_M \sqrt{\alpha_k} \partial |y_k^*| \quad \text{for all } k = 0, 1, \dots, K. \quad (33)$$

By the definition of $y^{(r)}$, we always have $x^{(r)}, y^{(r)}$ satisfy (33). Besides, by definition of $x^{(r)}$, we have

$$0 \in \rho \sum_{k=0}^K \left(x^{(r)} + y_k^{(r-1)} - c_k \right) + \lambda_M \partial |x^{(r)}|. \quad (34)$$

Thus, when

$$\rho \sum_{k=0}^K \left(y_k^{(r)} - y_k^{(r-1)} \right) = 0,$$

we will have $x^{(r)}, y^{(r)}$ satisfy (34). Let $\epsilon^{\text{abs}} > 0$ be an absolute tolerance and $\epsilon^{\text{rel}} > 0$ be a relative tolerance, we then stop at iteration r if

$$r^{\text{sub}} = \rho \left| \sum_{k=0}^K \left(y_k^{(r)} - y_k^{(r-1)} \right) \right| \leq (K+1)\epsilon^{\text{abs}} + \epsilon^{\text{rel}} \max \left\{ \sum_{k=0}^K |y_k^{(r)}|, \sum_{k=0}^K |y_k^{(r-1)}| \right\} := \epsilon^{\text{sub}}.$$

The final ADMM algorithm and the algorithm to solve the subproblem are summarized in Algorithm 1 and Algorithm 2.

For the proximal gradient descent algorithm to solve D-Trace loss objective (6), the stopping criterion is introduced as below, and the detailed algorithm is described in Algorithm 3.

Stopping criterion for proximal gradient. If $\Gamma^{(t)}$ is the solution for (6), the optimization criterion requires that

$$0 \in \nabla L_D(\Gamma^{(t)}) + \lambda_{\Gamma}^{(k)} \cdot \partial |\Gamma^{(t)}|_1.$$

Note that by the definition of $\Gamma^{(t)}$ in (11), we have

$$0 \in \Gamma^{(t)} - \left(\Gamma^{(t-1)} - \eta \nabla L_D \left(\Gamma^{(t-1)} \right) \right) + \eta \lambda_{\Gamma}^{(k)} \cdot \partial \left| \Gamma^{(t)} \right|_1, \quad (35)$$

which implies that

$$0 \in \nabla L_D(\Gamma^{(t)}) + \lambda_{\Gamma}^{(k)} \cdot \partial \left| \Gamma^{(t)} \right|_1 + \frac{1}{\eta} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) - \left[\nabla L_D \left(\Gamma^{(t)} \right) - \nabla L_D \left(\Gamma^{(t-1)} \right) \right].$$

Thus, when

$$\begin{aligned} r^D &= \left\| \frac{1}{\eta} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) - \left[\nabla L_D \left(\Gamma^{(t)} \right) - \nabla L_D \left(\Gamma^{(t-1)} \right) \right] \right\|_{\mathbf{F}} \\ &= \left\| \frac{1}{\eta} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) - \frac{1}{2} \widehat{\Sigma}^{(0)} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) \widehat{\Sigma}^{(k)} - \frac{1}{2} \widehat{\Sigma}^{(k)} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) \widehat{\Sigma}^{(0)} \right\|_{\mathbf{F}} \end{aligned}$$

is close to zero, we then have (35) hold approximately. Therefore, given an absolute tolerance $\epsilon^{\text{abs}} > 0$ and a relative tolerance $\epsilon^{\text{rel}} > 0$, we then stop at iteration t if

$$r^D \leq \epsilon^{\text{abs}} d + \epsilon^{\text{rel}} \times \max \left\{ \frac{1}{\eta} \left\| \Gamma^{(t)} - \Gamma^{(t-1)} \right\|_{\mathbf{F}}, \left\| \frac{1}{2} \widehat{\Sigma}^{(0)} \left(\Gamma^{(t)} - \Gamma^{(t-1)} \right) \widehat{\Sigma}^{(k)} \right\|_{\mathbf{F}} \right\}.$$

Algorithm 1 ADMM for Trans-MT-GLasso

- 1: **Input:** $\{\hat{\Sigma}^{(k)}\}_{k=0}^K$, λ_M , ρ , ϵ^{abs} , ϵ^{rel} and $\{\alpha_k\}_{k=0}^K$.
 - 2: **Initialize:** Let $\Omega^{(0)} = I_d$, $\Gamma_k^{(0)} = 0$ and $Z_k^{(0)} = I_d$ for all $0 \leq k \leq K$. Let $r^{\text{pri}} = r^{\text{dual}} = \infty$ and $\epsilon^{\text{pri}} = \epsilon^{\text{dual}} = 0$. Let $t = 0$.
 - 3: **while** $r^{\text{pri}} > \epsilon^{\text{pri}}$ or $r^{\text{dual}} > \epsilon^{\text{dual}}$ **do**
 - 4: $t \leftarrow t + 1$.
 - 5: **for** $k = 0, 1, \dots, K$ **do**
 - 6: Let

$$\tilde{C}_k^{(t-1)} = -Z_k^{(t-1)} + \left(\Omega^{(t-1)} + \Gamma_k^{(t-1)} \right) - (\alpha_k/\rho) \hat{\Sigma}^{(k)},$$
 and compute the eigenvalue decomposition of $\rho \tilde{C}_k^{(t-1)}$ as

$$\rho \tilde{C}_k^{(t-1)} = U \Lambda U^\top, \quad \Lambda = \text{diag} \left(\{\lambda_i\}_{i=1}^d \right).$$
 - 7: Let

$$\Omega_k^{(t)} = U \text{diag} \left(\left\{ \frac{\lambda_i + \sqrt{\lambda_i^2 + 4\rho\alpha_k}}{2\rho} \right\}_{i=1}^d \right) U^\top.$$
 - 8: Let

$$\check{C}_k^{(t)} = \Omega_k^{(t)} + Z_k^{(t-1)}.$$
 - 9: **end for**
 - 10: Solve

$$\Omega_{jl}^{(t)}, \left\{ \Gamma_{k,jl}^{(t)} \right\}_{k=1}^K = \mathcal{S} \left(\left\{ \check{C}_{k,jl}^{(t)} \right\}_{k=1}^K \right) \quad \text{for all } 1 \leq j, l \leq d.$$
 by Algorithm 1.
 - 11: Let

$$Z_k^{(t)} = Z_k^{(t-1)} + \rho \left(\Omega_k^{(t)} - \Omega^{(t)} - \Gamma_k^{(t)} \right) \text{ for all } 0 \leq k \leq K.$$
 - 12: Let

$$r^{\text{pri}} = \left(\sum_{k=0}^K \left\| \Omega_k^{(t)} - \left(\Omega^{(t)} + \Gamma_k^{(t)} \right) \right\|_{\text{F}}^2 \right)^{\frac{1}{2}},$$

$$r^{\text{dual}} = \rho \left(\sum_{k=0}^K \left\| \left(\Omega^{(t)} + \Gamma_k^{(t)} \right) - \left(\Omega^{(t-1)} + \Gamma_k^{(t-1)} \right) \right\|_{\text{F}}^2 \right)^{\frac{1}{2}},$$
 and

$$\epsilon^{\text{pri}} = \epsilon^{\text{abs}} d \sqrt{K+1} + \epsilon^{\text{rel}} \max \left\{ \left(\sum_{k=0}^K \left\| \Omega_k^{(t)} \right\|_{\text{F}}^2 \right)^{\frac{1}{2}}, \left(\sum_{k=0}^K \left\| \Omega^{(t)} + \Gamma_k^{(t)} \right\|_{\text{F}}^2 \right)^{\frac{1}{2}} \right\},$$

$$\epsilon^{\text{dual}} = \epsilon^{\text{abs}} d \sqrt{K+1} + \epsilon^{\text{rel}} \left(\sum_{k=0}^K \left\| Z_k^{(t)} \right\|_{\text{F}}^2 \right)^{\frac{1}{2}}.$$
 - 13: **end while**
 - 14: **Output:** $\check{\Omega}^{(k)} = \Omega^{(t)} + \Gamma_k^{(t)}$ for all $k = 0, 1, \dots, K$.
-

Algorithm 2 Solver of sub problem (30)

1: **Input:** $c = (c_0, c_1, \dots, c_K)$; Initial $x^{(0)}$ and $\{y_k^{(0)}\}_{k=0}^K$; λ_M , ρ and $\{\alpha_k\}_{k=0}^K$.
2: **Initialize:** $r^{\text{sub}} = \infty$ and $\epsilon^{\text{sub}} = 0$. Define $\text{ST}_\lambda(\cdot)$ as in (31). Let $r = 0$.
3: **while** $r^{\text{sub}} > \epsilon^{\text{sub}}$ **do**
4: $r \leftarrow r + 1$.
5: Let

$$x^{(r)} = \text{ST}_{\lambda_M/(\rho(K+1))} \left(\frac{1}{K+1} \sum_{k=0}^K (c_k - y_k^{(r-1)}) \right).$$

6: Let

$$y_k^{(r)} = \text{ST}_{\lambda_M \sqrt{\alpha_k}/\rho} (c_k - x^{(r)}) \quad \text{for all } 0 \leq k \leq K.$$

7: Let

$$r^{\text{sub}} = \rho \left| \sum_{k=0}^K (y_k^{(r)} - y_k^{(r-1)}) \right|,$$

$$\epsilon^{\text{sub}} = (K+1)\epsilon^{\text{abs}} + \epsilon^{\text{rel}} \rho \max \left\{ \sum_{k=0}^K |y_k^{(r)}|, \sum_{k=0}^K |y_k^{(r-1)}| \right\}.$$

8: **end while**

9: **Output:** $x^{(r)}$ and $\{y_k^{(r)}\}_{k=0}^K$.

Algorithm 3 Proximal Gradient Method for D-Trace Loss

1: **Input:** $\hat{\Sigma}^{(0)}, \hat{\Sigma}^{(k)}, \lambda_\Gamma^{(k)}, \eta, \epsilon^{\text{abs}}$ and ϵ^{rel} .
2: **Initialize:** $\Gamma^{(0)} = 0$, $r^D = \infty$, $\epsilon^D = 0$ and $t = 0$.
3: **while** $r^D > \epsilon^D$ **do**
4: $t \leftarrow t + 1$
5: Let

$$A^{(t-1)} = \Gamma^{(t-1)} - \eta \nabla L_D (\Gamma^{(t-1)})$$

$$= \Gamma^{(t-1)} - \eta \left\{ \frac{1}{2} (\hat{\Sigma}^{(k)} \Gamma^{(t-1)} \hat{\Sigma}^{(0)} + \hat{\Sigma}^{(0)} \Gamma^{(t-1)} \hat{\Sigma}^{(k)}) - (\hat{\Sigma}^{(0)} - \hat{\Sigma}^{(k)}) \right\}.$$

6: Let

$$\Gamma_{jl}^{(t)} = \left[|A_{jl}^{(t-1)}| - \lambda_\Gamma^{(k)} \eta \right]_+ \cdot A_{jl}^{(t-1)} / |A_{jl}^{(t-1)}|, \quad 1 \leq j, l \leq d.$$

7: Let

$$r^D = \left\| \frac{1}{\eta} (\Gamma^{(t)} - \Gamma^{(t-1)}) - \frac{1}{2} \hat{\Sigma}^{(0)} (\Gamma^{(t)} - \Gamma^{(t-1)}) \hat{\Sigma}^{(k)} - \frac{1}{2} \hat{\Sigma}^{(k)} (\Gamma^{(t)} - \Gamma^{(t-1)}) \hat{\Sigma}^{(0)} \right\|_{\text{F}},$$

and

$$\epsilon^D = \epsilon^{\text{abs}} d + \epsilon^{\text{rel}} \times \max \left\{ \frac{1}{\eta} \left\| \Gamma^{(t)} - \Gamma^{(t-1)} \right\|_{\text{F}}, \left\| \frac{1}{2} \hat{\Sigma}^{(0)} (\Gamma^{(t)} - \Gamma^{(t-1)}) \hat{\Sigma}^{(k)} \right\|_{\text{F}} \right\}.$$

8: **end while**

9: **Output:** $\hat{\Psi}^{(k)} = \Gamma^{(t)}$.

M Minimax Optimal Rate for Differential Network Estimation

Differential network estimation aims to estimate the difference between two precision matrices without first estimating the individual precision matrices. Although existing studies focus on providing upper bounds for this problem (Zhao et al., 2014; Yuan et al., 2017; Zhao et al., 2022), there is no known matching lower bound, making the minimax optimal rate an open question. As a byproduct of our analysis, in this section, we provide a minimax optimal rate for differential network estimation problem under certain conditions. To the best of our knowledge, this is the first minimax optimal guarantee towards this direction.

We start by formulating the problem setup. Note that we reintroduce some notation used in this section to make it self-contained, and one should not confuse it with the notation used in the other parts of the paper. Suppose that we have n_X i.i.d. samples $X_1, \dots, X_{n_X} \sim N(0, \Omega_X^{-1})$ and n_Y i.i.d. samples $Y_1, \dots, Y_{n_Y} \sim N(0, \Omega_Y^{-1})$. Let $\Delta := \Omega_Y - \Omega_X$ be the differential network between Ω_X and Ω_Y . Our goal is to use samples from two populations to estimate Δ . In addition, we assume that Ω_X and Ω_Y belong to the following parameter space.

$$(\Omega_X, \Omega_Y) \in \mathcal{M} := \{ \Omega_X, \Omega_Y \in \mathbb{S}^{d \times d} : c_1 \leq \gamma_{\min}(\Omega_X) \leq \gamma_{\max}(\Omega_X) \leq c_2, \\ c_1 \leq \gamma_{\min}(\Omega_Y) \leq \gamma_{\max}(\Omega_Y) \leq c_2, |\Delta|_0 \leq h, |\Delta|_1 \leq C_\Gamma \}, \quad (36)$$

where $c_1, c_2, C_\Gamma > 0$ are positive universal constants. The parameter space defined in (36) requires that the smallest eigenvalues and the largest eigenvalues of Ω_X and Ω_Y are lower bounded and upper bounded, respectively. Besides, we also require that the sparsity level of the differential network is bounded by h , and the L_1 norm of the differential network is

bounded by universal constant.

We first present the minimax lower bound for the above estimation problem. We state the result in the following theorem.

Theorem 1. *Assume that*

$$d \geq 4h, h \log d \geq 8 \log 3, \frac{h \log d}{n} \leq \min \{2, 8(1 - c_1)^2, 8(1 - c_2)^2\}, h \sqrt{\frac{\log d}{n}} \leq 4C_{\Gamma}. \quad (37)$$

We then have

$$\min_{\hat{\Delta}} \sup_{(\Omega_X, \Omega_Y) \in \mathcal{M}} \mathbb{E} \left[\left\| \hat{\Delta} - \Delta \right\|_{\text{F}}^2 \right] \gtrsim \frac{h \log d}{\min \{n_X, n_Y\}}$$

Proof. Let $\Omega_X = I_d$, note that

$$(I_d, \Omega_Y) \in \mathcal{M} \Leftrightarrow \Omega_Y \in \mathcal{G}_2,$$

where $\mathcal{G}_2$ is defined in (22). Now that samples from population X are useless for estimating Ω_Y , and we can only rely on samples from population Y to estimate Ω_Y . By Lemma 3, we then have

$$\min_{\hat{\Delta}} \sup_{(\Omega_X, \Omega_Y) \in \mathcal{M}} \mathbb{E} \left[\left\| \hat{\Delta} - \Delta \right\|_{\text{F}}^2 \right] \geq \min_{\hat{\Omega}_Y} \sup_{\Omega_Y \in \mathcal{G}_2} \mathbb{E} \left[\left\| \hat{\Omega}_Y - \Omega_Y \right\|_{\text{F}}^2 \right] \gtrsim \frac{h \log d}{n_Y}.$$

Similarly, we can show that

$$\min_{\hat{\Delta}} \sup_{(\Omega_X, \Omega_Y) \in \mathcal{M}} \mathbb{E} \left[\left\| \hat{\Delta} - \Delta \right\|_{\text{F}}^2 \right] \gtrsim \frac{h \log d}{n_X}.$$

Combine the above two inequalities, we have the final result. $\square$

Next, we derive the matching upper bound. Let $\hat{\Delta}$ be the D-Trace loss estimator defined in (6), and $\check{\Delta}(\tau)$ be the truncated version of $\hat{\Delta}$ as defined in (38). Following directly from Theorem 6, we then have the following theorem.

Theorem 2. Assume that $C_\Gamma \leq d^{\tau_3}$, where C_Γ is the same universal constant used in (36).

Let

$$\lambda_\Gamma \asymp \sqrt{\frac{\log d}{\min\{n_X, n_Y\}}},$$

where λ_Γ is the penalization parameter of D-Trace loss. Then for any $\tau \geq \tau_3$, we have

$$\sup_{(\Omega_X, \Omega_Y) \in \mathcal{M}} \mathbb{E} \left[\|\check{\Delta}(\tau) - \Delta\|_F^2 \right] \lesssim \frac{h \log d}{\min\{n_X, n_Y\}}.$$

Combine Theorem 1 and Theorem 2, we then have the final main result of this section.

Theorem 3. Assume that the conditions of Theorem 1 and Theorem 2 hold. We have

$$\min_{\hat{\Delta}} \sup_{(\Omega_X, \Omega_Y) \in \mathcal{M}} \mathbb{E} \left[\|\hat{\Delta} - \Delta\|_F^2 \right] \asymp \frac{h \log d}{\min\{n_X, n_Y\}}.$$

N Upper Bound for Expected Error

In this section, we develop theoretical guarantees for the expected error. We start with the analysis of the Trans-MT-Glasso. The subsequent theorem provides the expected error measured in the Frobenius norm.

Theorem 4. Suppose that Assumption 1 and Assumption 2 hold. Assume that $2(K+2) \leq d^{\tau_1}$ and $N \leq d^{\tau_2}$, for some universal constants $\tau_1, \tau_2 > 0$. In addition, assume that

$$\frac{\log d}{\min_{0 \leq k \leq K} n_k} \lesssim 1.$$

Let $\bar{n} = N/(K+1)$, $M_{\text{op}} \geq M_\Omega$ and $M_{\text{op}} = O(1)$. If $\lambda_M \asymp \sqrt{\log d/N}$, we have

$$\mathbb{E} \left[\sum_{k=0}^K \alpha_k \|\hat{\Omega}_k - \Omega^{(k)}\|_F^2 \right] \lesssim \left(\frac{s}{N} + \frac{h}{\bar{n}} \right) \log d.$$

Refer to the proof in Appendix O. The rate described in Theorem 4 consists of two parts. The first part, which is of the order $(s \log d)/N$, refers to the estimation of the shared component. The second part, of the order $(h \log d)/\bar{n}$, relates to the estimation of

the individual components.

As discussed in Section 5.2, the differential network estimate $\hat{\Psi}^{(k)}$ is considered the result of a black-box algorithm, with the presumption that its estimation errors are appropriately controlled. Let $\mathcal{B}_F(R) := \{A \in \mathbb{R}^{d \times d} : \|A\|_F \leq R\}$ be the ball with radius R . For $\tau > 0$ and $0 \leq k \leq K$, we define

$$\hat{\Psi}_{\text{proj}}^{(k)}(\tau) \in \arg \min_{\Gamma \in \mathcal{B}_F(d^\tau)} \left\| \Gamma - \hat{\Psi}^{(k)} \right\|_F. \quad (38)$$

$\hat{\Psi}_{\text{proj}}^{(k)}(\tau)$ is utilized solely for theoretical reasons. Suppose that by choosing τ appropriately, we have

$$\mathbb{E} \left[\left\| \hat{\Psi}_{\text{proj}}^{(k)}(\tau) - \Psi^{(k)} \right\|_F^2 \right] \lesssim \bar{g}_F^{(k)}(n_0, n_k, d, h, M_\Gamma) := \bar{g}_F^{(k)} \quad \text{for all } 0 \leq k \leq K. \quad (39)$$

We define

$$\hat{\Omega}_{\text{proj}}^{(0)}(\tau) = \sum_{k=0}^K \alpha_k \left(\check{\Omega}^{(k)} - \hat{\Psi}_{\text{proj}}^{(k)}(\tau) \right), \quad (40)$$

$\hat{\Psi}_{\text{proj}}^{(k)}(\tau)$ is defined in (38). Combining (39) with Theorem 4 yields the following theorem with expected error upper bound for Trans-Glasso.

Theorem 5. *If the conditions of Theorem 4 are satisfied and τ is chosen so that (39) holds, we have*

$$\mathbb{E} \left[\left\| \hat{\Omega}_{\text{proj}}^{(0)}(\tau) - \Omega^{(0)} \right\|_F^2 \right] \lesssim \left(\frac{s}{N} + \frac{h}{n} \right) \log d + \sum_{k=0}^K \alpha_k \bar{g}_F^{(k)}.$$

We then characterize $\bar{g}_F^{(k)}$ in the case where the D-Trace loss estimator in (6) is used.

Recall that $\hat{\Psi}_{\text{proj}}^{(k)}(\tau)$ is defined by (39).

Theorem 6. *Suppose that Assumption 1 and Assumption 2 hold. Assume that $2(K+1) \leq d^{\tau_1}$, $N \leq d^{\tau_2}$ and $M_\Gamma \leq d^{\tau_3}$ for some universal constants $\tau_1, \tau_2, \tau_3 > 0$. Let*

$$\lambda_\Gamma^{(k)} = M_\Gamma \sqrt{\frac{\log d}{\min \{n_k, n_0\}}}.$$

Then, for any $\tau \geq \tau_3$, we have

$$\mathbb{E} \left[\left\| \hat{\Psi}_{\text{proj}}^{(k)}(\tau) - \Psi^{(k)} \right\|_{\text{F}}^2 \right] \lesssim \frac{M_{\Gamma}^2 h \log d}{\min \{n_k, n_0\}}.$$

Proof. Note that $\|A\|_{\text{F}} \leq |A|_1$ for any matrix A . Thus we have $\|\Psi^{(k)}\|_{\text{F}} \leq \|\Psi^{(k)}\|_1 \leq 2M_{\Gamma}$.

The rest of the proof is similar to the proof of Theorem 4. $\square$

By Theorem 6, we have

$$\bar{g}_{\text{F}}^{(k)} = \frac{M_{\Gamma}^2 h \log d}{\min \{n_k, n_0\}}$$

for D-Trace loss estimator. Plugging the above results into Theorem 5, we then have the following corollary.

Corollary 2. *Let $\hat{\Omega}^{(0)}$ be obtained by Trans-Glasso (7) with the D-Trace loss estimator used in Step 1. Suppose that Assumption 1 and Assumption 2 hold, and that the conditions in Theorem 4 and Theorem 6 are satisfied. If $2(K+2) \leq d^{\tau_1}$, $N \leq d^{\tau_2}$, $M_{\Gamma} \leq d^{\tau_3}$ for some universal constants $\tau_1, \tau_2, \tau_3 > 0$, and*

$$\lambda_M = \sqrt{\frac{\log d}{N}}, \quad \lambda_{\Gamma}^{(k)} = M_{\Gamma} \sqrt{\frac{\log d}{\min \{n_k, n_0\}}} \quad \text{for all } k \in [K],$$

then for any $\tau \geq \tau_3$, we have

$$\mathbb{E} \left[\left\| \hat{\Omega}_{\text{proj}}^{(0)}(\tau) - \Omega^{(0)} \right\|_{\text{F}}^2 \right] \lesssim \left(\frac{s}{N} + (1 + M_{\Gamma}^2) \cdot \frac{h}{\bar{n}} + M_{\Gamma}^2 \cdot \frac{h}{n_0} \right) \log d. \quad (41)$$

The estimation error in (41) is comprised of three parts: shared component estimation, individual component estimation, and differential network estimation. If $\bar{n} \geq n_0$ and M_{Γ} is bounded by a universal constant, the error scales as $\frac{s \log d}{N} + \frac{h \log d}{n_0}$. When using only target samples, the lowest error rate achievable is $\frac{(s+h) \log d}{n_0}$ as stated in Lemma 2. Therefore, if $N \gg n_0$, the error rate can be significantly reduced compared to the optimal rate obtained with only the target samples. Moreover, as demonstrated in Section 5.3, the rate $\frac{s \log d}{N} + \frac{h \log d}{n_0}$

is minimax optimal under certain conditions.

O Proof of Theorem 4

Note that for any matrix $A \in \mathbb{R}^{d \times d}$, we have $\|A\|_F \leq \sqrt{d} \|A\|_2$. Since by our assumptions that $\|\Omega^{(k)}\|_2 = O(1)$ and $\|\check{\Omega}^{(k)}\|_2 = O(1)$, we thus have

$$\|\Omega^{(k)}\|_F = O\left(d^{\frac{1}{2}}\right) \quad \text{and} \quad \|\check{\Omega}^{(k)}\|_F = O\left(d^{\frac{1}{2}}\right).$$

For $\delta \in (0, 1]$, let

$$\lambda_M \geq C_1 C_3 \sqrt{\frac{\log(2(K+2)d^2/\delta)}{2N}},$$

where $C_1 = 160$ and $C_3 = M_\Sigma$. Besides, by the proof of Theorem 1, when

$$\frac{\log(2(K+2)d^2/\delta)}{\min_{0 \leq k \leq K} n_k} \leq \frac{1}{2},$$

then by (19), we have $\mathbb{P}\{\mathbb{G}(\lambda_M)\} \geq 1 - \delta$; or equivalently, we have $\mathbb{P}\{\bar{\mathbb{G}}(\lambda_M)\} \leq \delta$, where $\bar{\mathbb{G}}(\lambda_M)$ denotes the event that $\mathbb{G}(\lambda_M)$ does not hold.

Recall that by assumption we have $2(K+2) \leq d^{\tau_1}$. Let $\delta = d^{-\tau'}$, where τ' will be specified later, then by letting

$$\lambda_M = C_1 C_3 \sqrt{\frac{(\tau' + \tau_1 + 2) \log d}{2N}}, \tag{42}$$

$$\frac{(\tau' + \tau_1 + 2) \log d}{\min_{0 \leq k \leq K} n_k} \leq \frac{1}{2}, \tag{43}$$

we have $\mathbb{P}\{\bar{\mathbb{G}}(\lambda_M)\} \leq d^{-\tau'}$. Besides, by Section G.3, when (42)–(43) are true, $\mathbb{G}(\lambda_M)$ then implies that

$$\begin{aligned} \sum_{k=0}^K \alpha_k \|\check{\Omega}^{(k)} - \Omega^{(k)}\|_F^2 &\leq \frac{9(s + (K+1)h) \lambda_M^2}{4\kappa^2} \\ &\leq \frac{9C_1^2 C_3^2 (\tau' + \tau_1 + 2)}{8} \left(\frac{s}{N} + \frac{h}{n} \right) \log d. \end{aligned}$$

Note that

$$\begin{aligned} & \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \right] \\ &= \mathbb{P} \{ \mathbb{G}(\lambda_{\text{M}}) \} \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \mathbb{G}(\lambda_{\text{M}}) \right] + \mathbb{P} \{ \bar{\mathbb{G}}(\lambda_{\text{M}}) \} \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \bar{\mathbb{G}}(\lambda_{\text{M}}) \right]. \end{aligned} \quad (44)$$

Given (42)–(43) are true, we have

$$\mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \mathbb{G}(\lambda_{\text{M}}) \right] \leq \frac{9C_1^2 C_3^2 (\tau' + \tau_1 + 2)}{8} \left(\frac{s}{N} + \frac{h}{n} \right) \log d. \quad (45)$$

Besides, since $\|\check{\Omega}^{(k)}\|_{\text{F}}, \|\Omega^{(k)}\|_{\text{F}} = O(d^{\frac{1}{2}})$ for all $0 \leq k \leq K$, we have

$$\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \leq C' d,$$

for some constant $C' > 0$, and thus we have

$$\mathbb{P} \{ \bar{\mathbb{G}}(\lambda_{\text{M}}) \} \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \bar{\mathbb{G}}(\lambda_{\text{M}}) \right] \leq C' d \cdot d^{-\tau'} = C' d^{-(\tau'-1)}.$$

By Assumption that we have $N \leq d^{\tau_2}$ where $\tau_2 > 0$, then when we choose τ' such that

$$\tau' \geq \tau_2 + 1 + \frac{\log(8C'/9C_1^2 C_3^2 (\tau' + \tau_1 + 2))}{\log d},$$

we then have

$$\begin{aligned} C' d^{-(\tau'-1)} &\leq \frac{9C_1^2 C_3^2 (\tau' + \tau_1 + 2)}{8} \cdot \frac{1}{N} \\ &\leq \frac{9C_1^2 C_3^2 (\tau' + \tau_1 + 2)}{8} \left(\frac{s}{N} + \frac{h}{n} \right) \log d, \end{aligned}$$

which then implies that

$$\mathbb{P} \{ \bar{\mathbb{G}}(\lambda_{\text{M}}) \} \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \bar{\mathbb{G}}(\lambda_{\text{M}}) \right] \leq \mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_{\text{F}}^2 \middle| \mathbb{G}(\lambda_{\text{M}}) \right].$$

Combine the above inequality with (44) and (45), we finally have

$$\begin{aligned}
\mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_F^2 \right] &\leq 2\mathbb{E} \left[\sum_{k=0}^K \alpha_k \left\| \check{\Omega}^{(k)} - \Omega^{(k)} \right\|_F^2 \middle| \mathbb{G}(\lambda_M) \right] \\
&\leq \frac{18C_1^2 C_3^2 (\tau' + \tau_1 + 2)}{8} \left(\frac{s}{N} + \frac{h}{n} \right) \log d \\
&\lesssim \left(\frac{s}{N} + \frac{h}{n} \right) \log d.
\end{aligned}$$

P Graph Recovery Guarantee for Gaussian Graphical Models

While this paper primarily focuses on precision matrix estimation, it is also of great interest to explore the implications of Trans-Glasso for graph recovery in Gaussian graphical models (GGMs), given the close relationship between the precision matrix and the underlying graph structure. In this section, we introduce a graph estimation procedure based on thresholding the output of Trans-Glasso, establish a theoretical guarantee for its accuracy, and demonstrate its advantage over graph estimators that rely solely on target data.

Let $E^{(0)} = \left[E_{jl}^{(0)} \right]_{1 \leq j, l \leq d} \in \{0, 1\}^{d \times d}$ denote the adjacency matrix of the undirected Gaussian graphical model for the target data. Specifically, for any $j \neq l$, we set $E_{jl}^{(0)} = 1$ if nodes j and l are connected, and $E_{jl}^{(0)} = 0$ otherwise. By definition, $E_{jj}^{(0)} = 0$ for all $1 \leq j \leq d$. Given the well-established connection between GGMs and the precision matrix (Lauritzen, 1996), we have, for any $j \neq l$,

$$E_{jl}^{(0)} = \begin{cases} 1 & \text{if } \Omega_{jl}^{(0)} \neq 0, \\ 0 & \text{if } \Omega_{jl}^{(0)} = 0. \end{cases}$$

We propose a graph estimator based on thresholding the output of Trans-Glasso, $\hat{\Omega}^{(0)}$. Specifically, let $\hat{E}^{(0)} = \left[\hat{E}_{jl}^{(0)} \right]_{1 \leq j, l \leq d} \in \{0, 1\}^{d \times d}$ denote the estimated adjacency matrix. We

set $\widehat{E}_{jj}^{(0)} = 0$ for all $1 \leq j \leq d$, and for any $j \neq l$, define

$$\widehat{E}_{jl}^{(0)} = \begin{cases} 1 & \text{if } |\widehat{\Omega}_{jl}^{(0)}| > \epsilon_n, \\ 0 & \text{if } |\widehat{\Omega}_{jl}^{(0)}| \leq \epsilon_n, \end{cases}$$

where ϵ_n is a thresholding parameter.

We measure the graph estimation error using the Hamming distance between $E^{(0)}$ and $\widehat{E}^{(0)}$, defined as

$$\text{Err}(\widehat{E}^{(0)}) := \frac{1}{2} \sum_{1 \leq j, l \leq d} \mathbf{1} \left\{ \widehat{E}_{jl}^{(0)} \neq E_{jl}^{(0)} \right\}.$$

Additionally, let $\omega_{\min} = \min_{(j,l): E_{jl}^{(0)}=1} |\Omega_{jl}^{(0)}| > 0$ denote the minimum signal strength in the graphical model. The following deterministic bound then holds for $\text{Err}(\widehat{E}^{(0)})$.

Proposition 1. *If $0 < \epsilon_n \leq \frac{1}{2}\omega_{\min}$, then*

$$\text{Err}(\widehat{E}^{(0)}) \leq \frac{1}{2\epsilon_n^2} \left\| \widehat{\Omega}^{(0)} - \Omega^{(0)} \right\|_F^2.$$

Moreover, if the right-hand side is smaller than 1, perfect graph recovery is achieved, i.e., $\widehat{E}^{(0)} = E^{(0)}$.

Proof. Let

$$S_{\text{I}} = \left\{ (j, l) : E_{jl}^{(0)} = 0, \widehat{E}_{jl}^{(0)} = 1 \right\} \quad \text{and} \quad S_{\text{II}} = \left\{ (j, l) : E_{jl}^{(0)} = 1, \widehat{E}_{jl}^{(0)} = 0 \right\}$$

denote the sets of type-I error and type-II error. We then have

$$\text{Err}(\widehat{E}^{(0)}) = \frac{1}{2} \sum_{1 \leq j, l \leq d} \mathbf{1} \left\{ \widehat{E}_{jl}^{(0)} \neq E_{jl}^{(0)} \right\} = \frac{1}{2} (|S_{\text{I}}| + |S_{\text{II}}|).$$

Note that for any $(j, l) \in S_{\text{I}}$, we have

$$\left| \widehat{\Omega}_{jl}^{(0)} - \Omega_{jl}^{(0)} \right| = \left| \widehat{\Omega}_{jl}^{(0)} \right| > \epsilon_n. \tag{46}$$

Besides, for any $(j, l) \in S_{\text{II}}$, we have

$$\left| \hat{\Omega}_{jl}^{(0)} - \Omega_{jl}^{(0)} \right| \geq \left| \Omega_{jl}^{(0)} \right| - \left| \hat{\Omega}_{jl}^{(0)} \right| \geq \left| \Omega_{jl}^{(0)} \right| - \epsilon_n \geq \omega_{\min} - \epsilon_n \geq \epsilon_n, \quad (47)$$

where the last inequality follows from the assumption that $\epsilon_n \leq \frac{1}{2}\omega_{\min}$.

Combining (46) and (47), we have

$$\begin{aligned} \left\| \hat{\Omega}^{(0)} - \Omega^{(0)} \right\|_{\text{F}}^2 &\geq \sum_{(j,l) \in S_{\text{I}}} \left(\hat{\Omega}_{jl}^{(0)} - \Omega_{jl}^{(0)} \right)^2 + \sum_{(j,l) \in S_{\text{II}}} \left(\hat{\Omega}_{jl}^{(0)} - \Omega_{jl}^{(0)} \right)^2 \\ &\geq \epsilon_n^2 (|S_{\text{I}}| + |S_{\text{II}}|) = 2\epsilon_n^2 \cdot \text{Err} \left(\hat{E}^{(0)} \right), \end{aligned}$$

which then implies the final result. $\square$

To illustrate the advantage of Trans-Glasso in graph recovery over methods relying solely on target data, we consider a special case of the parameter space defined in (21), where $\Omega_{jj}^{(0)} = 1$ for all $1 \leq j \leq d$. In this scenario, the sample-optimal graph recovery algorithm proposed by Misra et al. (2020), which uses only target data, requires

$$n_0 \gg (s + h) \log d / \omega_{\min}^2 \quad (48)$$

to achieve perfect graph recovery.

By Proposition 1 and Corollary 2, we set $\epsilon_n \asymp \omega_{\min}$ with $\epsilon_n \leq \omega_{\min}/2$. When the conditions of Corollary 2 hold and $\bar{n} \geq n_0$ with $K \leq d^\tau$ for some constant $\tau > 0$, perfect graph recovery requires only

$$n_0 \gg h \log d / \omega_{\min}^2 \quad \text{and} \quad N \gg s \log d / \omega_{\min}^2. \quad (49)$$

Comparing (48) and (49), the target sample size requirement for n_0 under Trans-Glasso is much smaller than that of target-only methods when $s \gg h$, illustrating the benefit of transfer learning in Gaussian graphical model estimation.

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